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Work Package H

Earth Observation

Deliverable H3

Final technical report

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Acronyms and abbreviations

AES	Agri-Environmental Support
ANB	Agency for Nature and Forests
ANN	Artificial Neural Network classifier
AOI	Area of Interest
ARMA	Agency Restructuring and Modernisation of Agriculture
BAG	Basisregistratie Adressen en Gebouwen
BDtopo	Topographic database
CAP	Common Agricultural Policy
CESBIO	Center for the Study of the Biosphere from Space
CLC	CORINE Land Cover
CNES	National Centre for Space Studies in France
CRS	Coordinate Reference System
CTM	Chemical Transport Models
CS	Case Study
DGFiP	General Direction of Public Finances
DHMV	Digital Hoogtemodel Vlaanderen
DMSP	The Defense Meteorological Program
DNB	Day/Night Band
DT	Decision Tree
EEA	European Environment Agency
EO	Earth Observation
EODaS	Earth Observation Data Science
ESA	European Space Agency
EVI	Enhanced Vegetation Index
FCNN	Fully Convolutional Neural Networks
FN	False Negatives
FP	False Positives
GHS- SMOD	Global Human Settlement Model grid
GHSL	The Global Human Settlement Layer
GIS	Geographic Information System
GIAB	Generic Information Architecture for Big Data
GRD	Ground Range Detected product
GSIM	Generic Statistical Information Model
IACS	Integrated Administration and Control System
ICNF	Institute of the Conservation of Nature and Forests
IGN	National Institute of Geography and Forestry
ILVO	Research Institute for Agriculture, Fisheries and Food
IMCC	Imperviousness Classified Change
INSPIRE	Infrastructure for Spatial Information in Europe
IW	Interferometric Wide
KIA	Kappa Index of Agreement
KNN	K- Nearest Neighbour
LAEA	Lambert Azimuthal Equal Area Projection
LEDAPS	The Landsat Ecosystem Disturbance Adaptive Processing System
LiDAR	Light Detection and Ranging
LC	Land Cover
LPIS	Land Parcel Identification System
LST	Land surface temperature
LUCAS	Land Use and Coverage Area frame Survey
MMU	Minimum Mapping Unit
MODIS	Moderate Resolution Imaging Spectroradiometer
NDMI	Normalized Difference Moisture Index
NDTI	Normalized Difference Turbidity Index

NDVI	normalized difference vegetation index
NetCDF	Network Common Data Format
NIR	Near Infrared
NRC/LC	National Reference Centres Land Cover
NSS	National Statistical System
NUTS	Nomenclature of Territorial Units for Statistics
OA	Overall Accuracy
OSM	Open Street Map
OSO	CES Occupation des sols
OVAM	Flemish Public Waste Agency
PACA	Provence Alpes Côte d'Azur
RF	Random Forest
S1	Sentinel 1
S2	Sentinel 2
SAR	Synthetic Aperture Radar
SDG	Sustainable Development Goal
SDI	Spatial Data Infrastructure
SegNet	Segmentation Network
SRTM	Shuttle Radar Topography Mission
Statbel	Belgian statistical office
SVM	Supported Vector Machine
SWIR	Shortwave Infrared
TIFF	Tagged Image File Format
TN	True Negatives
TOA	Top of atmosphere
TP	True Positives
UAV	Unmanned Aerial Vehicle
UV	Ultraviolet
VEA	Flemish Energy Agency
VH	vertical transmitted and horizontal received signal
VI	Vegetation Index
VIIRS	Visible Infrared Imaging Radiometer Suite
VITO	Vlaams Instituut voor Technologisch Onderzoek
VLM	Flemish Land Agency
VNIR	Visible and near- infrared
VV	Vertical transmitted and vertical received signal
WMS	Web Map Service
WPH	Work Package H
WUP	Weighted Urban Proliferation

1. Introduction

This pilot project aims to support areal statistics with Earth Observation (EO) data. Project results in experimental statistics using remote sensing data. From the technological point of view, the WPH uses new methods like machine learning algorithms for image analysis. Within this project, the usefulness and practical usage of EO data in order to fill the gap between statistical and geographical information named "geospatial breakdown" is presented. The main objectives of WPH are implemented by the execution of different case studies (CS) divided into thematic tasks: agriculture, build-up area, land cover, settlements, enumeration areas and forestry. The overview of thematic tasks and cases studies is presented in Figure 1.1.

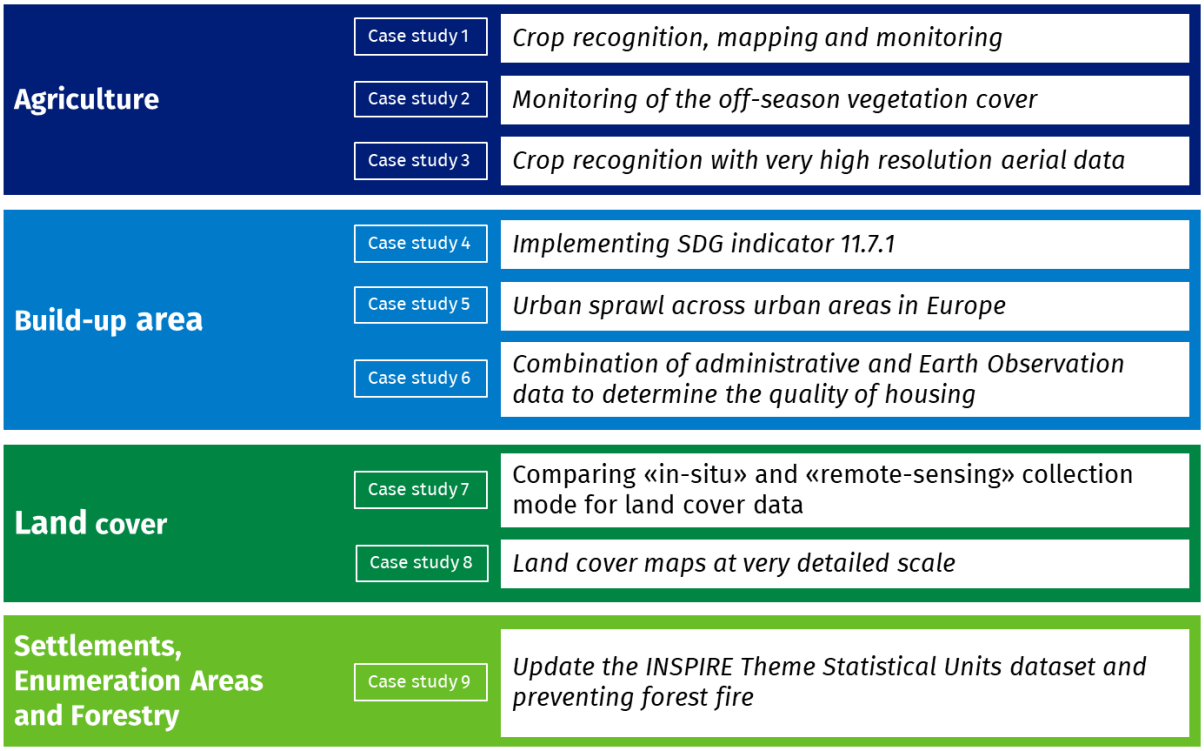


Figure 1.1 Overview of thematic tasks and cases studies.

This document starts from executive summaries of each case study (Section 2) to briefly introduce readers to the topics of the work, while the whole description of each case study is presented in Interim Technical Report (Deliverable H1) and Report on results and activities executed by case studies (Deliverable H2) available on webpage:

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WPH_Milestones_and_deliverables

The next sections of this document focus on: evaluation of big data sources and definition of possible statistical products from examined big data sources (Section 3); business architecture suitable for big data processing (Section 4); methodological framework and its general relation to quality, IT, and metadata aspect (Section 5); quality framework with the protection of privacy and confidentiality and other legal issues (Section 6); IT infrastructure (Section 7).

2. Executive summaries of cases studies

2.1. Case study 1 - Crop recognition, mapping and monitoring

The main purpose of the "Agriculture - Crop recognition, mapping and monitoring" case study is to use Sentinel 1 and Sentinel-2 satellites for agricultural crops mapping and area estimates in Northern Europe conditions. Based on previous experience from ESSnet Big Data I (2016-2018) the main crops identification is possible, but the accuracy sometimes can be questionable and should be improved. This case study aims to verify and improve the plausibility of the crop recognition methodology. Detailed analysis and discussion about the accuracy of the classifier have been taken. The different classifiers were tested. For the pilot study, the Warmia and Mazury voivodship (Poland) was selected as a region of interest. The remote sensing and geodata used in this case study are:

- time-series (October 2017 – September 2018) of Sentinel-1 GRD images (multi-looked and projected to ground range using an Earth ellipsoid model) acquired in Interferometric Wide swath mode,
- collection of Sentinel-2A/B VNIR/SWIR images for the same period,
- cadastral parcels limits from ARMA agency (Poland),
- agricultural plots borders from General Geographic Geodatabase (GUGIK),
- information on crops declared by farmers (ARMA),
- information on crop types and/or species from field visits (Central Statistical Office).

The gathered data were prepared properly from the source format to the ingested format. The collection of remote sensing dataset was pre-processed for analyses and classifications. The pre-processing consisted of radiometric and geometric transformations of Sentinel-1A/B images leading to the elaboration of orthorectified sigma nought maps. In the case of Sentinel-2 images, the key pre-processing step was image mosaicking and cloud masking. The pre-processing was done using open source software – SNAP 7.0 (ESA). The geodata were reformatted, the databases were re-shaped and necessary information was extracted using QGIS 2.18 software. The geodata ingestion provides the cadastral parcels limits, which is facilitating image segmentation, objects aggregation and validation.

In order to select the subsets of images, the most suitable for object-based classification the time series analysis has been done. Merging temporal and polarimetric features of the crops permitted to extract the subsets returning maximum separability of crops and highly reduced data volume. This task was very important and critical for the final results. It gave the answer which crops can be effectively recognized and mapped using Sentinel-1A/B very long time series.

In order to achieve the goals, object-based classification including machine learning algorithms were performed. Key elements of this task were:

- testing mean shift segmentation algorithm parameters for calculating homogeneous areas (segments) on Sentinel-1 and Sentinel-2 pre-processed data with open source OrfeoToolbox software (CNES),
- testing machine learning algorithms (support vector machine classifier, decision tree classifier, artificial neural network classifier, random forest classifier, KNN classifier) parameters in the context of obtaining the best accuracy for crop recognition with open source OrfeoToolbox software (CNES)

At the end, the results obtained in the classification processing were validated. The validation included: extraction of validation samples from ARMA and in situ data, computing confusion matrix for each classifier, comparing results to the official statistics.

The main purpose of crop recognition and mapping with high accuracy, using time series of radar Sentinel-1data supported by optical Sentinel-2 data, was achieved. Crop monitoring is the step just after the crop recognition and mapping. The presented case study shows a huge potential of use EO data in agricultural statistical production due to easy data access, global coverage and reliable results. One of the limitations can be complex processing and 10 m spatial resolution causes small parcels elimination. Despite this global estimation of crop groups area is possible. The distinction between some crops is hard due to the similarity in a plant structure (i.e cereals) taking into account the method of obtaining data (i.e satellite radar). The general conclusion is that the presented methodology can support agricultural statistics in further projects including crop yields and growth models.

2.2. Case study 2 - Monitoring of the off-season vegetation cover

The general objective of this task was to monitor the off-season vegetation cover of agricultural soils in high-latitude agricultural systems. The draft implementing regulation of the integrated farm statistics (IFS) data collection in 2023 includes data on soil cover on arable land in winter. Large-scale information on off-season vegetation cover will enhance the estimation of soil erosion and nutrient loads to the water bodies and carbon balance of the soil. The statistics on off-season soil cover relate to the UN's Sustainable Development Goals, Indicator 2.4.1: Proportion of agricultural area under productive and sustainable agriculture, as land management practices closely relate to sustainability.

Specifically, a pilot case of a machine learning based classifier by merging Earth observation data, administrative data and agro-meteorological data was established. For predictive modelling, radar satellite images from the European Space Agency's Copernicus Sentinel-1 mission and optical satellite data from the Copernicus Sentinel-2 mission were utilized. In addition, open data on agro-meteorological conditions and soil properties were used. As reference data, farmers' declarations from the Integrated Administration and Control System (IACS) were used. As an outcome, the remote sensing based classifier provides information e.g. on the amount of vegetation vs. bare soil coverage in a timely fashion in relatively good accuracy. Thus, IACS data can stand as a sole source of reference data for land use classification. Although, having in-situ data e.g. from on-the-spot checks data by IACS agency, would further improve the accuracy assessment of a classifier.

Pre-processed 11-day mosaics of Sentinel-1 data and Sentinel-2 based indices covering the whole of Finland were utilized. The products have been processed at the National Satellite Data Center at the Finnish Meteorological Institute. The data sets are publicly available on the Finnish Geospatial Platform under a Creative Commons license. The Finnish Geospatial Platform is a joint effort of the public administration, the Ministry of Agriculture and Forestry being the responsible party. The platform collects spatial data from various providers and makes them openly available to users. The aim of the platform is to consolidate spatial data services provided by the public administration, to improve data-based decision-making and save public administration costs, e.g. by enabling the efficient maintenance of data resources and removing overlapping activities. Access to pre-processed satellite data proved to be of utmost importance for statistics production, as remote sensing based technologies would have needed specialized expertise and IT resources. It was able to establish a pilot with basic level knowledge in remote sensing and have most of the focus on data analysis.

2.3. Case study 3 - Crop recognition with very high-resolution aerial data

The use of satellite data with an intermediate resolution (10 m for Sentinel-1/2) for crop recognition in agricultural statistics is not adequate in regions where the size of parcels is relatively small and the diversity of crops, often of high added value, is relatively high. An alternative approach, in this case, would be to estimate the acreage for each of the various crops growing on small parcels using aerial photography data with a higher resolution.

The case study on this detailed crop recognition needs to address two issues:

1. Data availability: are aerial photography data available for official statistics, with characteristics and conditions of use allowing them to be realistically exploitable?
2. Testing in practice whether high-resolution aerial photography data make crop recognition possible at a more diversified, detailed and small-scale level, using machine learning algorithms.

The second question can only be tackled after the first one has been answered positively and the required data have become available. As this proved impossible within the project time frame, Case study 3 had to be limited to assessing actual and potential data availability.

Belgium is a federal state where competencies linked to the territory, among them agriculture and land use, are exercised at the level of Belgium's three regions (Flanders, Wallonia, Brussels). The Belgian statistical office (Statbel) opted to focus on Flanders, partnering with Statistics Flanders (acting as 'unofficial' non-refunded WPH partner). Flanders has 6.6 million inhabitants for a surface area of 13,500 km², resulting in a high population density of almost 500 persons per km². Consequently, its agricultural area (7,425 km² or 55%) consists of a great number of relatively small parcels which may vary considerably as to the crops under cultivation.

Within the Flemish administration, the GIS unit/Geoloket (Geocounter) of the department Agriculture and Fisheries runs the EU agricultural surveys also using satellite data and machine learning, and several research institutions study the potential of imagery for land use and agriculture purposes: EODaS (Earth Observation Data Science) Programme, VITO (Vlaams Instituut voor Technologisch Onderzoek, Flemish Institute for Technological Research), ILVO (Instituut voor Landbouw- en Visserijonderzoek, Research Institute for Agriculture, Fisheries and Food). These entities were contacted with the purpose of inventorying satellite and aerial photography datasets and assessing current and future practices for obtaining, creating and using them.

The general conclusion from contacts with EODaS, VITO and Geoloket is that satellite data of intermediate resolution (Sentinel via Terrascope data platform) are freely available at the required high frequency, while high-resolution aerial photography data can only be provided annually (25 cm - in winter, moreover), tri-annually (40 cm) or 10-yearly (10 cm), a temporal resolution unsuitable for detailed crop recognition.

VITO, however, additionally also creates or buys remote sensing data (satellite, airplane manned or unmanned, drone) financed by clients in particular projects.

Data availability: Producing aerial photography datasets

A possible alternative to accessing freely available high-resolution aerial images of sufficient frequency might be to produce these via airplane or drone (UAV, Unmanned Aerial Vehicle), possibly partnering with other public agencies or research institutes to lower the cost.

Focusing on drones rather than planes which seem too expensive, Case study 3 has established the required specifications of aerial photography datasets as to area to be covered, resolution, period and frequency (number of iterations), altitude, speed, drone type, coverage per flight session, angle and image spectral type(s). Based on this, an approximate unit price per covered area was estimated, and the practical and legal possibilities and constraints were investigated.

As a general conclusion, aerial photography datasets seem too expensive to be used exclusively, but they could be integrated into a three-tier mixed-mode statistical production consisting of satellite data analysed via AI algorithms, trained and supplemented by survey and administrative data, as the 'normal' and usual way to recognise crops, aerial photography data for the limited and clearly defined areas with small parcels and a wide range of high-value-added crops, and surveying in those areas where the use of drones is not allowed.

Finally, another possible approach is buying detailed satellite data, at the appropriate resolution of 30 cm from private companies.

Because the first subtask, assessing the data situation, came to the conclusion that high-resolution aerial photography data are not readily available with the required frequency, the next foreseen stage of testing machine learning methods to analyse the data could not be commenced. It was, for the time being, limited to a first look into the data science capabilities of the various partners to address the research question (Statbel, Statistics Flanders, VITO, Geocounter).

Additionally, a commented overview of reports, documents and webpages on similar applications and projects by the organisations and units contacted in the course of Case study 3 was compiled.

Due to the lack of freely available aerial photography, the second research question on testing them for crop recognition on small parcels could not be addressed. Creating or buying aerial photography data for limited areas seems to be a viable approach when combined with satellite and survey data. The next steps are testing the usability of aerial photography data in practice, setting up a business model for their use in official statistics, and developing partnerships with producers and/or other users to do so.

2.4. Case study 4 - Implementing SDG indicator 11.7.1

Aim of this case study is to implement the UN methodology to compute the SDG indicator 11.7.1 "average share of the build-up area of cities that is open space for public use for all, by sex, age and persons with disabilities" using Earth Observation data. Moreover, the final product can be refined by: i) distinguishing among the "open space for public use" between the streets, the green open public spaces and others open public spaces, ii) giving the proportion of people (by sex and age) who live within the cities who have access to these open public spaces (except streets) meaning they are located at less than 400m from the space. The combined information from OSO land cover map (derived based on Sentinel-2 optical images) and geodata from Open Street Map, national topographic database (Bdtopo) and cadastral plan is a base of the research. The whole process is divided into two steps. The first step is to derive the cities boundaries from the OSO land cover map. The second step is to delineate within those boundaries the areas which are open for public use. The cities boundaries were obtained by pixel clustering for 31 urban units in France with more than 200 000 inhabitants. Open public spaces and streets delineations within cities are more tricky as no sources can provide accurately all the open public spaces. For this reason, the area types of interest should be clearly stated and data from various sources should be combined. Bdtopo contains a specific layer tagged as "open public spaces" with parks and squares, but not includes green corridors. Additionally, some public spaces are represented by single points or

line, not a continuous area. These spaces were tagged as fictional and they were converted to area geometry combing information from Open Street Map and cadastral plan. Delineation of open public spaces within cities allowed for the calculation i) the proportion of open public spaces covered by BDTOPO and OSM data simultaneously ii) proportion of BDTOPO open public spaces covered by "non fictional" elements and iii) proportion of the city area covered by 1500m buffers around open public spaces. Based on these three indicators quality score for open public spaces was calculated and normalised by the highest score reached among all cities (which is the city of Lyon with a quality score of 0,57). There is a small positive correlation between the quality score and the normalized quality score: cities with the lowest value of the indicator also has the lowest values for the quality score. But some cities with a low quality score has a high value of the indicator like Le Mans or Le Havre. Moreover, most cities have a low quality score which means that there is great uncertainty over the final results and a need for deeper analysis in the future.

2.5. Case study 5 - Urban sprawl across urban areas in Europe

This case study concerns a method of calculation UN SDG indicator 11.3.1: "Ratio of land consumption rate to population growth rate" to estimate urban sprawl by using Earth Observation data. The study focused on two parts of the Netherlands, comparing a highly urbanized area near Amsterdam known as the Randstad with a lesser urbanized area around the city of Maastricht in the south of the Netherlands. The first area has increasing numbers of inhabitants and is estimated to provide housing to 50% of the Dutch population, while the second area has seen a decrease in inhabitants. By analysing the development of these two distinct areas over time, the authors hoped to provide insight into the concept of urban sprawl for the Netherlands, and derive techniques that can be applied to different regions throughout Europe.

To this cause, data from several Earth Observation sources, as well as several sources of administrative information providing a in-detail overview of the land use and building characteristics in the Netherlands were identified. Satellite imagery from three different providers was analysed. The first data source are MODIS Terra and Aqua at 250-500m resolution and can be obtained from 2004 to the present, with a frequency of one image every sixteen days. The second data source is Landsat 8 providing data with a 30m resolution with a similar frequency as the MODIS data. The last data source is Sentinel-2, providing data with a 10m resolution. All of these satellites deliver information of surface reflectance which can be recalculated into many spectral indices like normalized difference vegetation index (NDVI). As potential administrative data sources the Dutch building register, the Dutch register on soil use classification, the CORINE land cover data and the Global Human Settlement Layer were identified.

Two datasets were prepared using the available data sources: a single-scene dataset and a multi-scene dataset. In the case of the single-scene dataset, a Landsat 8 image was taken covering most of the Netherlands. The machine learning algorithms were trained on a single scene for July 2018, and then evaluated on the same geographical area, but a different time period spanning from July 2015 to June 2018. In the case of the multi-scene dataset, all Landsat 8 data since its launch until 2018 were considered, but focused on two distinct geographical areas. One of the areas was used for training and testing the machine learning algorithm, while the other one was used a blind test area to estimate the extent to which the algorithm was generalizing to other geographical areas not seen before. All data for the regions of interest are projected into a regular 30m grid and zonal statistics for each of the spectral bands are computed. Moreover, a land use cover category is assigned to each grid cell using the CORINE data. At the end of the data pre-processing, each grid cell in the region of interest has satellite measurement

statistics assigned, used as input for the machine learning algorithm, as well as land cover labels, used as target label when training the machine learning algorithm.

Before training the machine learning algorithm, the pre-processed data were explored by analysing random sub-selections to identify patterns. It became clear that agricultural areas dominate the regions of interest and that unbalances in the dataset, when training, have to be taken into account. The distributions of three different remote sensing indices (NDVI, MNDBI, and NDBaI) were explored on several areas of interest, along with their corresponding label distributions from the CORINE dataset. From this exploration, it became clear that a combination of the three different remote sensing indices exhibits sufficiently different behaviour to discriminate between different types of land use cover. In addition to the aforementioned features, three additional spectral features describing greenness, wetness, and brightness were selected as input to the machine learning classifier.

After the data preparation and feature selection, a Random Forest Classifier was trained on the available data. To this cause, 80% of the dataset was used for training, the remaining 20% for testing. Moreover, the training was carried out using a 5-fold cross-validation to find optimum hyper-parameters and rule out random factors while training. Furthermore, because the distribution of land cover classes was unbalanced, class weights were assigned to each class during training. Training and evaluation were carried out in both the single and multi-scene settings described above, both leading to similar model performance. Overall accuracy was 0.89 for the best model on the test set, an overall accuracy of 0.83 for the blind test area. Recall, precision, and F1-score were also measured per category. While overall performance was satisfactory, the variations for these measures per land cover category were quite large. The model especially had difficulties identifying forest/semi natural areas and water bodies. However, the main goal of the model was distinguishing between build-up area and other areas, which the model seems to be able to do sufficiently.

Finally, a method for estimating urban sprawl using the weighted urban proliferation index (WUP), a metric evaluating the build-up area, its dispersion, and the uptake of built-up area per inhabitant was presented. The WUP was calculated and visualized for two cities in the Netherlands on a 100x100m grid: Den Haag, a highly populated city, and Maastricht, a lesser populated city. While a preliminary analysis shows promising results, further validation using domain experts is still needed.

2.6. Case study 6 - Combination of official statistics and Earth Observation data to determine the quality of life

The aim of this case study is to investigate the benefit of combining earth observation data with official statistics. By this, new research questions could be analysed; potentially on a deeper regional level than it was possible before. Precisely, this investigation is about the environmental quality of urban life. Due to its high spatial and temporal resolution, earth observation data can be used to measure certain aspects of environmental quality of life such as air quality, urban heat islands and urban green. By combining geocoded official statistical data and earth observation data, this topic can be addressed in a comprehensive way.

The German microcensus was geocoded for the first time in 2018. This creates the chance to link information about the surroundings concerning one explicit grid cell (e.g. a 100 x 100 metres grid cell) to the according geocoded household. To demonstrate this, the topic of environmental urban quality of life was chosen, because - as the literature review in this case study shows - the neighbourhood has a direct influence on the wellbeing of persons living in a household e.g. through the accessibility of green areas.

In a first step, this case study will identify the environmental quality of life aspects that are measured with remote sensing techniques. Secondly, these data will be linked to the geocoded households of the survey.

Different aspects concerning the environmental quality of life were identified through reviewing the quality of life initiatives, discussions with earth observation experts and a literature review on determining the quality of life using remote sensing data. Furthermore, a literature review of the identified topics (environmental surroundings indicators like temperature and vegetation) was conducted. Based on the identified aspects, possible data sources have been investigated, specifically for the identified topics together with further data sets that could be linked to a geocoded survey.

Official data stem from the German microcensus, which contains information about the socio-economic situation of a household. This information, merged with remote sensing data, can be used to determine differences in the quality of life within cities. The case study analyses the correlations of certain sets of variables that might indicate specific environmental life conditions for certain socio-economic groups. The analysed remote sensing data is gained from the Sentinel-2 satellites with a 10 x 10 metres spatial resolution and from the Landsat 8 satellite with a 30 x 30 metres spatial resolution. The data of one day in July 2018 from these satellites are gathered and processed to link them to the 100 x 100 metres grid cell of the German microcensus. The results show a moderate correlation of vegetation and temperature as is expected. However, the correlations between the environmental and socio-economic factors hardly exist for the analysed test area.

Concluding, this case study shows that even if the concrete correlation results do not exactly match the theoretical hypothesis from literature, there is a great potential for new and more granular data analysis regarding the aspect of spatial resolution. Further data on the environmental living conditions (e.g. air quality and noise pollution) would enhance this kind of analysis for an evidence-based democratic decision-making process concerning diverse socio-economic groups of the population. This can be achieved through the combination of earth observation data with official statistics.

2.7. Case study 7 - Comparing «in-situ» and «remote-sensing» collection mode for land cover data

The Teruti survey

Teruti is the french statistical survey on land cover and land use. It has been carried out every year since 1982 by the Office of Statistics and Prospective analysis (SSP) in the Ministry of Agriculture and Food.

Teruti is an area-frame statistical survey. The sampling frame is a square grid of points spaced 250 m apart covering the entire territory. Each point in the grid constitutes the statistical unit of the survey, a portion of territory delimited by a circle of 3 m diameter on the ground and centred on the geographical coordinates of the point. For about 80% of the sample frame points, the land cover is directly imputed from geolocated databases covering water surfaces, buildings, roads, agricultural parcels and forest massifs. For the remaining 20% of the territory, a sample of 70,000 points is selected each year for field observation by an investigator. By aggregating imputed data and field observations, Teruti provides a statistical breakdown of the national territory into 13 land cover categories up to the departmental geographical level (NUTS3).

The OSO land cover map : a fully automatic production by the Cesbio

Since 2016, the Scientific Expertise Centre on Biosphere (CESBIO) has carried out and disseminated an annual land cover map (OSO) in 23 categories over the whole french metropolitan territory. This map is automatically produced from images taken by the Sentinel2 remote sensing satellites dedicated to earth observation in the framework of the European Copernicus program. The 2 Sentinel2-A and 2-B satellites observe the entire earth's globe with a 290 km field of view and 5 days repetitiveness of images. Sentinel2 images have a resolution ranging from 10 m to 60 m according to the spectral bands corresponding to the different wavelengths of the light received by the sensor (reflectance). These images contain 13 spectral bands ranging from infrared to visible light in order to detect different types of land cover based on their spectral signature.

Iota^2 : the processing chain for satellite images classification

The CESBIO developed a processing chain for the classification of satellite images taking advantage of the high resolution and repetitiveness of Sentinel2 images. This Iota^2 chain implements a supervised classification method that assigns each pixel into one out of 23 land cover categories depending on the pixel's spectral features. The method allows processing large areas and large volumes of data, fully automatically and in a short time. The geo-localized databases from the National Geographic Institute, the geo-localized agricultural parcels from the CAP payments control system and Corine Land Cover are the main reference databases used for training the classification model. The classification algorithm used is Random Forest which leads to accurate classifications with less complexity and less processing time. The classification of the Sentinel2 images in the land cover categories is processed taking into account a partition of the territory by eco-climatic region and tiles of 110 km x 110 km. The overall accuracy index reaches 90% for the OSO 2019 map.

The land cover changes detection method

One of the main goals of the Teruti survey is to produce an accurate estimate of land cover changing areas between two surveys, at a relatively fine geographical level (NUTS3 department). Since land cover changes are a rare phenomenon, one of the ways to improve the accuracy of indicators is to target field surveys on Teruti points likely to have evolved in order to observe as many as possible.

Therefore, CESBIO and SSP have experimentally developed a method for detecting changes in land cover at Teruti points by comparing annual OSO maps derived from satellite images. This method was developed in several stages. The first step was to build a harmonized land cover nomenclature between Teruti and OSO. Then new OSO maps have been recalculated from 2016 to 2019 taking this harmonised nomenclature into account. The interesting types of land cover changes or transitions were also selected from this nomenclature. After several setting tests, it was decided to select the land cover transitions to be observed from a direct comparison between the OSO 2017 map and the 2019 map, wherever the confidence of the pixel classification was greater than 50% for both years. To enhance the relevance of the detected transitions, the method allows to take into account not only the pixel of the Teruti point but also the neighbouring pixels.

First results and prospects

Over the french metropolitan area, this method of detection has identified more than 700,000 land cover changes among the 8.8 million points of the Teruti grid. About 10% of these changes are located on Teruti points which belong to the field-observation stratum. Within this 10%, the main transitions

identified are “deforestation” (21%), “agricultural intensification” (21%), “agricultural land abandonment” (19%).

In 2020, Teruti survey investigators return to the points already observed three years ago, in 2017. The Teruti 2020 field-collection will thus constitute the «ground truth» to improve and verify the relevance of the land cover changes detection method derived from the statistical processing of satellite images. For the Teruti 2021 and subsequent surveys, the land cover changes detected from the OSO maps could form a new stratum of the Teruti survey sample design.

2.8. Case study 8 - Land cover maps at very detailed scale

The Italian National Institute of Statistics (Istat) is currently investigating whether Deep Learning methods could be used to derive automated Land Cover estimates of satisfactory quality from Sentinel-2 satellite images. A prototype software system is being developed within the scope of this research. The present case study about “land cover maps” focuses on a very relevant, though quite specific, output artefact of the system. Deliverable H1 accounts for the foundations and early results of our project, whereas deliverable H2 critically discusses the limitations of our initial attempts and illustrates ongoing countermeasures.

An automatic Land Cover (LC) estimation system should be able to take as input a satellite image depicting a portion of territory, and to return as output a table of LC statistics. Although LC estimation is a quantification problem rather than a classification one, the system according to a pure ‘classify and count’ design was started implementing. The main driver of this initial design choice was to incorporate into our system a Convolutional Neural Network (CNN), so as to take advantage of its tremendous performance in image classification tasks. Without going into details, our classify-and-count design can be summarized as follows:

(1) Train a CNN to predict the LC class of a satellite image ‘tile’ (i.e. a small, fixed-size sub-image).
(2) Divide the satellite images covering a ‘target area’ (i.e. the territory for which LC statistics have to be computed) into tiles.
(3) Use the trained CNN to predict the LC class of all the tiles generated in step (2).
(4) Obtain LC statistics for the target area by simply computing the absolute or relative frequencies of predicted LC classes.

The EuroSAT dataset was adopted as a training set for our CNN. As CNN model, Google’s Inception-V3 architecture was selected, which next was customized and trained on the EuroSAT dataset. The adoption of EuroSAT as a training set has the following major consequences: (i) system of Istat can only produce LC estimates according to EuroSAT LC classification (only 10 classes, non-hierarchical), and (ii) LC predictions can only be obtained for tiles of size 64 x 64 pixels. Point (i) unfortunately prevents any direct comparison among the LC estimates produced by Istat system and those derived by flagship projects like CORINE and LUCAS, whose LC taxonomies are hierarchical and much richer than EuroSAT one.

After the release of Deliverable H1, the accuracy of the classify-and-count approach documented therein was carefully tested. The quality evaluation on sample input images in two ways was carried out. First, the automated LC estimates produced by Istat system with information available from CORINE and LUCAS (to this end devise tentative and approximate mappings between different classifications had to be done) was compared. Second, the corresponding automated LC maps were carefully examined visually. This analysis showed that Istat automated LC estimates had a remarkably good accuracy for most LC classes, except for a systematic upward bias affecting narrow linear structures like rivers

and highways. This overestimation issue turned out to be inextricably linked to the adoption of EuroSAT as a training set and to Istat tile-based classify-and-count approach. In fact, EuroSAT constrains Istat CNN to process tiles of size 64 x 64 pixels, whose surface area is quite big (about 41 hectares). But rivers and highways are narrow linear structures. Therefore, any highway fragment framed into a tile occupies just a rather small portion of the tile. Nonetheless, whenever the CNN correctly detects a highway in a tile, our system attributes the whole surface area of the tile to the 'Highway' class. This evidently leads to overestimation. The same happens, of course, when the 'River' class is concerned.

As reported in deliverable H2, Istat decided to address the overestimation issue affecting the original design of Istat system by integrating Istat tile-based CNN model with a proper image-segmentation algorithm. "Image-segmentation" is the task of localizing objects of interest within a digital image (e.g. rivers in our application scenario) by identifying their shape and/or contours. The segmentation process groups different image pixels together according to their features, and binds them to a common class label (e.g. 'River'). Among several solutions for image-segmentation proposed in the field of Deep Learning, the choice fell on the U-Net model for its intuitive architecture and because it succeeded in many real-world applications. The basic idea that guides the re-design and improvement of the system is simple:

(1) Train the U-Net to identify and reconstruct only rivers and highways. Use the trained U-Net to produce partial LC maps of input images (i.e. maps where only classes 'River' and 'Highway' are detected).

(2) Train our existing Inception-V3 CNN on all the other LC classes of the EuroSAT dataset, and let the old-fashion classify-and-count approach produce partial LC maps of input images (i.e. maps where classes 'River' and 'Highway' are no longer detected).

(3) Properly merge the partial LC maps produced in step (1) and (2).

(4) For all the pixels of the merged map where the U-Net detected either class 'River' or class 'Highway', trust the U-Net and neglect the predictions of the CNN.

The final output of steps (1) – (4) is an integrated, complete (and still automated) Land Cover map, encompassing all the original EuroSAT LC classes. Automated LC estimates can eventually be obtained from this integrated map by simply computing class frequencies across its pixels.

While the logic underpinning the proposed re-design and improvement of our system is simple, it comes with non-negligible costs in terms of data preparation. Indeed, in order to train the U-Net to perform the segmentation of rivers and highways, a suitable training set has to be constructed for each of the corresponding LC classes. In deliverable H2, authors focused on the 'River' LC class and only quickly hinted at ongoing developments for the 'Highway' class. A good segmentation training set for rivers requires (i) samples of diverse Sentinel-2 images representing portions of rivers of different sizes and shapes, along with (ii) the corresponding segmentation masks. These segmentation masks can be thought as binary images (e.g. black and white images) where each pixel value encodes the information about the presence (white pixel) or absence (black pixel) of a river in the pixel location. While the EuroSAT dataset offers plenty of European rivers images (2'500), the generation of the corresponding segmentation masks is, of course, a non-trivial task. The authors tackled this task by leveraging the auxiliary information on European water areas provided by Copernicus High Resolution Layers. This procedure entailed two phases. In the first, automatically build provisional seg-mentation masks for all the 2'500 river examples was done. In the second, these masks were evaluated by visual inspection, rejecting those whose correspondence to the input EuroSAT image was not accurate enough. This way, the 1500 validated segmentation masks univocally associated to EuroSAT 'River' examples were obtained.

Deliverable H2 provides (i) sample results of our U-Net trained on the 'River' segmentation dataset described above, as well as (ii) one example of a final (i.e. merged) LC map obtained for a large Sentinel-2 input image (the 'Pisa image'). The latter output was generated by reconciling and integrating the partial maps produced, for the same territory, by the Inception-V3 CNN and by the U-Net. As was expected and hoped for, the 'River' overestimation issue no longer affects the final LC map returned by our re-designed system.

Istat is currently using OpenStreetMap data as auxiliary information to create segmentation masks associated to EuroSAT 'Highway' images, along the lines of what was done for the 'River' class.

2.9. Case study 9 - First topic - Update the INSPIRE Theme Statistical Units dataset, namely the Settlements and Enumeration Areas

The objective of the first topic aims to update the Settlements and Enumeration Areas integrated in the INSPIRE Theme Statistical Units dataset by redefining the geography of the statistical subsections or census tracts to the next 2021 Census. The support by earth observation data related to built-up area and human presence is also expected. EO data allows identifying urban and rural dynamics in the last years thus requiring multi-temporal data and overcoming administrative and statistical boundaries.

This topic presents two methodologies: i) the first one based on land use Copernicus data (Global Human Settlement Layer and Imperviousness Classified Change) and spatial analysis techniques in local GIS environment ii) the second methodology took advantage of the night-time lights data and Impervious data available on the Google Earth Engine platformFirst methodology was a more direct approach compared to the second one due to the short processing time and small size data processing as well as complexity of the tools and process. The results are presented at municipality level (LAU 1) for mainland Portugal; ii) the second one based on other data sources – Global Artificial Imperviouss Area (GAIA) – Imperviousness surface data –, and night-time lights data (two satellites – DMSP-OLP and VIIRS). The results are presented for mainland Portugal and NUTS III Região de Coimbra - which comprises 19 municipalities (168 parishes – LAU 2), integrating the third largest metropolitan area in Portugal in terms of the resident population – by municipality, 1 km^2 grid and level 9 GSGF hexagrid.

Concerning the first methodology, the three datasets (GHS-SMOD 2015, GHS-BUILT Multi-temporal 1975-2014 and IMCC 2012-2015) were collected by tiles covering mainland Portugal in order to optimize time extraction. In GIS environment the selected tiles were merged to create a new raster and extracted from the outline of the test site (mainland Portugal). The raster data was converted to vector data keeping data structure in order to further operationalise spatial analysis. On the other hand, in the second methodology, no pre-processing was needed for the night-time light data and the GAIA imperviousness data since already pre-processed data are available on Google Earth platform. However the following vector format datasets required preparation, projection to WGS 1984 Web Mercator and upload to personal Google account: polygon boundaries of the Portuguese municipalities, the level 9 GSGF hexagrid of mainland Portugal and 1 km^2 grid for the study area.

In the first methodology two main tasks were executed regarding main data processing: i) selection of the Settlements and Enumeration Areas (in the national statistical framework are designated "localities") of 2011 census; and ii) selection of earth observation data based on classification (e.g. built-up area or urban area) and temporal criteria (e.g. time period). It is important to mention that the delimitation of the statistical units is based on both administrative and/or geographic boundaries, therefore

the built-up area is not a criterion for its definition. The summary statistics of the statistical units presented a multi-dimension variance - in the 2011 census more than 85% of the statistical units have less size than 1 km^2 - which is important to highlight when working with earth observation data at high resolution.

The second methodology took advantage of the night-time lights data and Impervious data available on the Google Earth Engine platform by getting indicators for the growth of urban areas after 2011 for mainland Portugal. In case of the Impervious data, it was also aim to have a local copy of the study area. Slightly different procedures were used for both data sources namely using query and classify operations in which the essential differences between them are: i) multiple images for each year exist for the night-time lights data, on the contrary, the GAIA data consist of only one image for all the years (in this image each pixel is classified with the year of transition from pervious to impervious); and ii) no value is associated to each pixel, only the classification of the year.

Regarding the analysis of the results, in the first methodology the main output includes both statistical units and datasets geographies, having the identification of matching and non-matching areas, that is to say, the area of statistical units that are within the grid cells as well as the area of grid cells that is outside the statistical units. Based on this spatial output several spatial analysis were operationalised highlighting the situation of overestimation on the summary statistics based on the spatial resolution of the grid cells (20 meters to 1 km) and the multi-dimension feature of the statistical units. In the second methodology, it was clear that with relatively little data processing it was possible to get data that could be easily associated with Portuguese geographic reference data: municipalities, 1 km^2 grid and level 9 GSGF hexagrid. The resulting CSV tables can be easily be downloaded for further data processing.

All the datasets presented the potential to develop territorial-based statistical products. The surface of artificial areas (built-up area or imperviousness land) can support a universal and standard territorial indicator under the National Statistical System (NSS) and disseminated by Statistics Portugal: population density. The population density of built-up areas by municipality (estimates of the resident population of 2018) for mainland Portugal was calculated and compared to the standard indicator identifying the territories with a higher concentration of population in the settlements (less urbanisation dispersion). On the other hand, the night-time lights data are only indicative for imperviousness land, however it is possible to include these data in a statistical product or statistical workflow. Two indicators were developed based on data from both satellites: i) Growth in night-time lights between 1995 and 2013 by municipality for mainland Portugal using DMSP-OLP data; and ii) Growth in night-time lights between 2014 and 2019 by HexaGRID using VIIRS data which has higher resolution. Additionally, the GAIA dataset also provides information about the change in the surface of artificial areas, thus a growth imperviousness area indicator (2012-2018) in relation to total area was develop by municipality and 1 km^2 grid.

The main advantages of this type of data are: i) the open and free access supported established international initiatives ensuring the regular production and viability of the data source; ii) the higher resolution enabling a multi-scale analysis and dissemination of statistics; and iii) multi-temporal data enabling spatial-temporal analysis and statistics.

2.10. Case study 9 – Second topic - Explore the possibility of studying the forest and the eucalyptus plantation and its impact in preventing forest fire

This work has two main objectives: i) assess the susceptibility of occurrence of forest fires in the NUTS III Região de Coimbra based on the history of the burned areas as a dependent variable (2000 to 2019)

and the 6 conditioning factors (altitude, slope, aspect, population density, distance to road network and land use); and ii) describe a model that allows identifying Eucalyptus forest areas in NUTS III Região de Coimbra using an artificial neuronal network, more specifically a convolutional neuronal network (CNN) applied to satellite images of the European Environment Agency (ESA) of the Copernicus earth observation programme, including Sentinel-1 and Sentinel-2 images with a 10 meter resolution.

The following sources and datasets were used: i) burnt areas (2000-2016; 2018-2019), mainly mapped based on satellite imagery and field survey; ii) European Elevation Model (2011); iii) road network (2011); iv) Population and Housing Census (2011); and v) Land Use Map (2018). Data pre-processing was based on the analysis and comparison of two statistical models: Information Value (relatively simple and evaluates quantitatively the degree of susceptibility) and Linear Regression (explores and models relationships between one or more explanatory or independent variables and a dependent variable); these both methods will produce an indicator to assess the susceptibility to forest fires resulting on a statistical product. These methods were operationalised taking into account a set of the dependent variable (burnt areas) and conditioning factors (altitude, slope, exposure of the slopes, distance to the road network, population density and land use). The set of conditioning factors is important since the modelling of the susceptibility of forest fires goes beyond the history of forest fires in which these conditioning factors are the most commonly used.

The spatial outputs of the conditioning factors in spatial data pre-processing and treatment were crossed with the dependent variable - burnt areas -, resulting in the spatial statistics regarding frequency (%), burnt area per class (km^2) and conditional probability (%). The conditional probability represents the probability of finding a certain dependent event, in this case, the forest fires associated with the presence of a factor (or class of this factor) in the study area. On the other hand, the calculation of conditional probabilities associated with the various classes of each of the variables aims to understand how these conditions the phenomenon studied as well as to define the relative importance of the variables.

The class of each conditioning factor with the highest Informative Value was identified - Informative Value scores of the factors for the construction of the susceptibility model -, in other words the variables with the highest influence on the distribution of burnt areas resulting on susceptibility map. For the occurrence of the forest fires for the study area by susceptibility level (IV > 1: high susceptibility; 0 < IV < 1: moderate susceptibility; -1 < IV < 0: reduced susceptibility; and VI < -1: very low susceptibility). Subsequently, the Linear Regression model was applied including all relevant conditioning factors (independent variables) and classes and resulting in two outputs. The first output of this model was the logistic regression coefficients for each conditioning factor indicating, in general terms, that the predisposition factors are significant for the construction of the model in which the exception was the exposure of the slopes factor (p-value - significance -greater than 0.005). The second output was the classification matrix which compares the classifications made by the model and the actual observations in the modelling and validation group presenting a value of 69.6% overall accuracy.

Concerning the eucalyptus forest identifying by image classification using Machine Learning methods, two Sentinel-2 satellite images with a minimum resolution of 10 meters were downloaded from the Copernicus Open Access Hub website. The reference date of the images is May 2018, as well as the reference year of the Land Use Map. Bands of radar Synthetic Aperture radar (SAR) of Sentinel-1, which can provide much information about land cover under all climatic conditions, were also used. In the first stage to train the model, the COS (Land Use and Land Cover Map) of 2018 were used in which each

polygon will be as a training sample that will feed the model. Classes of the COS were grouped to avoid working with a higher number of classes and various classes were included (not only the Eucalyptus Forest class) in order to the model better distinguish the Eucalyptus from the other occupation classes. Before creating the model, ArcGIS Pro requires image chips of training samples (small cut out image of the training image of the location of the training samples), therefore a directory with images, image labels, image statistics and definition model that references each image were created. Thus this definition model will be used to train the model generating thousands of image chips.

For the training process of the model was used the interface of ArcGIS Pro where only the number of epochs was configured, with the value 25 – the number of times that the image chips will be processed by the neuronal network -, as well as the batch parameter that defines the number of training samples processed simultaneously, which has been set to only 1, to avoid memory failure problems. From the several types of models available, the U-Net model (pixel classification) was used because it is one of the best know semantic image segmentation algorithms for land cover classification. Semantic segmentation is a process that associates a class to each pixel of an image, so that pixels with the same class share certain similar characteristics in the various image bands.

The result of the model training is a generated file of type Esri Model Definition (emd) that executes geoprocessing tools to detect or classify resources. A template information document is also created, including a chart illustrating training and validation losses and real/predicted comparisons. After training the model it is possible to perform the inference which is the process that allows extracting information from the image and identifying for each pixel the land class to which it belongs. This action was executed in the image with the reference date of May 2020, similar to the image of May 2018, with the same bands and from the infrared spectral regions, and the same resolution (20 meters). This modelling process produced a land cover map with the land cover classes for the study area.

The data from the Land Use Map (2010 to 2018) and the results obtained in this work lead us to believe that the Eucalyptus area in NUTS III Região de Coimbra has doubled since 2010 with 500 km^2 to almost 1200 km2, despite the decree of law nº77/2017 that aims to halt the expansion of the Eucalyptus. At the municipality level, there are already several municipalities with more than 50% of Eucalyptus area such as Mealhada, Miranda do Corvo and Penacova, compared to the values presented in 2010. For future research, it would be interesting to notice what land cover classes are being replaced by the eucalyptus forest thus to understand land use dynamics, namely in rural areas.

The datasets used in this action, in particular the European Digital Elevation Model and the Sentinel Images as well as the methodologies applied, have the potential to produce statistical indicators of the evolution of the area occupied by the Eucalyptus and a related indicator concerning the susceptibility of the forest to fire. The open access and high-resolution earth observation data with regular periodicity ensures the viability of the statistical products.

3. Evaluation of big data sources and definition of possible statistical products from examined big data sources.

The WPH concerns four thematic fields and is implemented by the execution of nine different case studies and therefore is related to different end products. The expected products of WPH can be grouped into three categories: statistical indicators, basic research and product supporting existing systems for statistical production (Table 3.1).

Table 3.1. Classification of expected products within WPH in reference to each case study.

Thematic field	Case study	Main specification of final product		
		Statistical indicator	Basic research (experimental and theoretical research) = analysis	Support/improvement of existing systems for statistical production
Agriculture	CS1	+		
	CS2	+		
	CS3	+		
Build-up area	CS4	+		
	CS5	+		
	CS6		+	
Land cover	CS7	+		+
	CS8	+		+
Settlements, Enumeration Areas and Forestry	CS9	+		+

Statistical indicators can be defined as the area under cultivation and cropped area (CS1, CS3), the proportion of agricultural area under vegetation cover in winter – SDG indicator 2.4.1 (CS2), the average share of the build-up area of cities that is open space for public use for all, by sex, age and persons with disabilities – SDG indicator 11.7.1 (CS4), Weighted Urban proliferation index (CS5), land change (CS7), indicators of the evolution of the area occupied by the Eucalyptus and a related indicator concerning the susceptibility of the forest to fire (CS9). Basic research is the output of the CS6. Through the combination of different data sources, a statistical analysis of how the urban quality of life differs between socio-economic groups can be conducted. The collected geographic data in CS6 is output as well, which can be used as a basis for further research and statistical production. In the case of the CS8 the research goal is to design and develop an automatic Land Cover (LC) estimation system. Such a system should be able to take as input a satellite image depicting a portion of territory and to return as output a table of LC statistics.

The Big Data used in WPH can be classified as geodata (data with spatial/geographic reference). Geodata gives information named as geoinformation, that can be used successfully in statistical production. Geodata comes in many different forms, such as

- Satellite images
- Aerial images/UAV images
- Other elaborations (maps, databases)
- Administrative data
- Surveys

Satellites are an essential data source in the context of Earth Observation applications within this work-package. However, additional geodata are needed to make the satellite data fully useful. These additional geodata may be called auxiliary, supporting, external, or ground truth data and they may come from the sources listed above.

The recognized and applied data to achieve the intended objectives of the WPH are listed in Table 3.2. A detailed description of accessing the data can be found in the document called H1 - Interim technical report

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/3/31/WPH_Deliverable_H1_Interim_technical_report_2019_11_05.pdf

Table 3.2 List of data sources.

	Data	Provider/Owner	Access
CS1	Sentinel -1, Sentinel-2	European Space Agency (ESA) / European Commission (EU)	Free /open
	Land Parcel Information System (LPIS)	Agency for Restructuring and Modernisation of Agriculture (ARMA)	Free /agreement
	Farmers declarations	Agency for Restructuring and Modernisation of Agriculture (ARMA)	Free /agreement
	Cadastral plan	Head Office of Land Surveying and Cartography (GUGIK)	Free /agreement
	In-situ measurement	Statistics Poland (GUS)	Free /own data of NSI
CS2	Sentinel-1, Sentinel-2	European Space Agency (ESA) / European Commission (EU)	Free /open
	Land Parcel Information System (LPIS)	Finnish Agency for Rural Affairs	Free /agreement
	farmers declaration on environmental commitments (from AES – Agri Environmental Support Scheme)	Finnish Agency for Rural Affairs	Free /agreement
	Meteorological data	Finnish Meteorological Institute	Free /open
	Finish Soil Database	Natural Resources Institute Finland (Luke)	Free /open
CS3	Aerial photography at an insufficient frequency	Flemish Institute for Technological Research (VITO)	Free /agreement
	Aerial or UAV (Unmanned Aerial Vehicle) photography	Private companies	Cost /limited
CS4	Land cover map (OSO)	Center for the Study of the Biosphere from Space (CESBIO)	Free /open
	Cadastral plan	General Directorate of Public Finances (DGFiP)	Free
	Topographic database (BD TOPO)	National Geographical Institute (IGN)	Free /agreement
	Open Street Map (OSM)	OpenStreetMap Foundation	Free /open
CS5	Sentinel-1,Sentinel-2	European Space Agency (ESA) / European Commission (EU)	Free /open
	Landsat 8, Terra/Aqua MODIS	NASA/USGS	Free /open
	Registry of addresses and buildings (BAG)	The Netherlands' Cadastre, Land Registry and Mapping Agency (Kadaster)	Free /agreement
	Soil Use Classification	Statistics Netherlands (CBS)	Free /own data of NSI
	Corine Land Cover (CLC 2006)	European Environment Agency (EEA) / European Commission (EU)	Free /open
	Global Human Settlement Layer (GHSL)	Joint Research Centre (JRC) / European Commission (EU)	Free /open
CS6	Sentinel-2, Sentinel-5p	European Space Agency (ESA) / European Commission (EU)	Free /open
	Landsat 8	NASA/USGS	Free /open
	Geodata: Digital Elevation Model (DEM) and Digital Surface Model	Federal Agency for Cartography and Geodesy (BKG)	Free /agreement

	(DSM), points of Interests (POI), House Coordinates (HK-DE/HU-DE)		
	Urban Atlas	European Environment Agency (EEA) / European Commission (EU)	Free /open
	Microcensus	Federal Statistical Office (Destatis)	Free /own data of NSI
CS7	Sentinel-2	European Space Agency (ESA) / European Commission (EU)	Free /open
	Land cover map (OSO)	Center for the Study of the Biosphere from Space (CESBIO)	Free /open
	TERUTI	Ministry of Agriculture	
CS8	Sentinel-2	European Space Agency (ESA) / European Commission (EU)	Free /open
	EuroSAT dataset	German Research Center for Artificial Intelligence	Free /open
CS9 topic 1	Global Human Settlement Built-up area grid (GHS-BUILT)	Joint Research Centre (JRC) / European Commission (EU)	Free /open
	Global Human Settlement Model grid (GHS-SMOD)	Joint Research Centre (JRC) / European Commission (EU)	Free /open
	Imperviousness Classified Change (IMCC)	European Environment Agency (EEA) / European Commission (EU)	Free /open
	Night-time Lights Time Series (DMSP OLS)	NOAA's National Geophysical Data Center - US Air Force Weather Agency	Free /open
	Stray Light Corrected Night time Day/Night Band Composites Version 1 (VIIRS)	Earth Observation Group, NOAA National Geophysical Data Center	Free /open
	Tsinghua FROM-GLC year of change to impervious surface (GAIA)	Tsinghua University	Free /open
CS9 topic 2	Sentinel -1, Sentinel-2	European Space Agency (ESA) / European Commission (EU)	Free /open
	European Digital Elevation Model (EU-DEM)	European Environment Agency (EEA) / European Commission (EU)	Free /open
	Land Use Map 2018	Directorate-General of the Territory	Free /open
	Burnt areas	Institute for Conservation of Nature and Forests (ICNF) (also resource provider)	Free /open

The main providers of the used satellite data are European Space Agency (ESA) and National Aeronautics and Space Administration (NASA) cooperated with the U.S. Geological Survey. The Earth observation data delivered by the Sentinel, Landsat, Terra, Aster satellites are considered as public goods and are available to users on a free, full and open basis. These data cover the whole Earth, but at the cost of the lower spatial resolution (max. 10 m). The high-resolution satellite data are available, but they are still expensive, which makes a significant obstacle to their acquisition. The high-resolution satellite data haven't been used in the case studies (Table 3.2). Aerial images are also difficult to obtain due to costs. Those that are freely available cannot be used for detailed crop recognition via machine learning in spite of the adequate resolution, because they are recorded at an insufficient frequency (annually and tri-annually, and even 10-yearly for the highest-resolution data) to constitute the time series needed for analysis which should probably be at least weekly in the critical periods. Furthermore, some datasets cannot be used for crop recognition at all due to a recording time unfit for this purpose (during winter). If there are no high-resolution aerial images of sufficient frequency are freely available for the purpose of detailed crop recognition, one possible alternative might be for statistical institutes themselves to produce the required datasets via airplane or drone (UAV, Unmanned Aerial Vehicle). The cost of this could be lowered by partnering with public agencies or research institutes interested in aerial photography datasets for their own testing or operational objectives in the domains of agriculture or GIS.

The rest of the identified sources provide free data. In the case of national institutions sometimes an agreement is needed. There are no visible legal or ethical problems to access the data.

Since the listed data sources are free and open there are no seen limitations to the amount of data that can be accessed. In case of Sentinel data, only limitation can be a maximum of 2 concurrent downloads per user allowed in order to ensure a download capacity for all users. Data are stored on online archive of at least the latest year of products for immediate download via HTTP. Older data and least used data are removed from online archives and they are on demand. Processing of farmers' declaration by paying agencies takes much time and usually, this data can be achieved just after vegetation season. Other limitations are related to the cost of aerial imageries.

4. Business architecture suitable for big data processing.

The aim of WPH is to use Earth Observation data for producing statistics in different thematic areas e.g agriculture, build-up areas, land cover, settlements, enumeration areas and forestry. This chapter will describe the application and information architecture of WPH according to the Big Data Reference Architecture and Layers (BREAL) https://ec.europa.eu/eurostat/cros/sites/crosportal/files/WPF_Deliverable_F1_BREAL_Big_Data_REference_Architecture_and_Layers_v.03012020.pdf :

- Application architecture: representation linking the business processes and functions, external application services and application components and services views.
- Information architecture: description of the information according to the raw data layer, the convergence layer and the statistical layer representations.
- Operational model.

4.1. Application architecture

As mentioned, WPH consists of nine case studies and various products. These products can be created using similar architectures. The application architecture is divided into three horizontal parts, respectively for business processes or functions, external application services and application components. Every business function triggers the next one. Every business function is served by one or more application services. Every application service is realized by one or more application components.

Figure 4.1 presents harmonised architecture for statistical production in reference to agriculture (CS1, CS2, CS3), build-up areas (CS5), land cover (CS8) and forestry (CS9). Despite the thematic diversity, a common denominator is the use of raw EO data to compute statistical indicators by applying image classification with machine learning algorithms. Thus presented architecture can be applied as well to issues raised in CS6 about quality urban life.

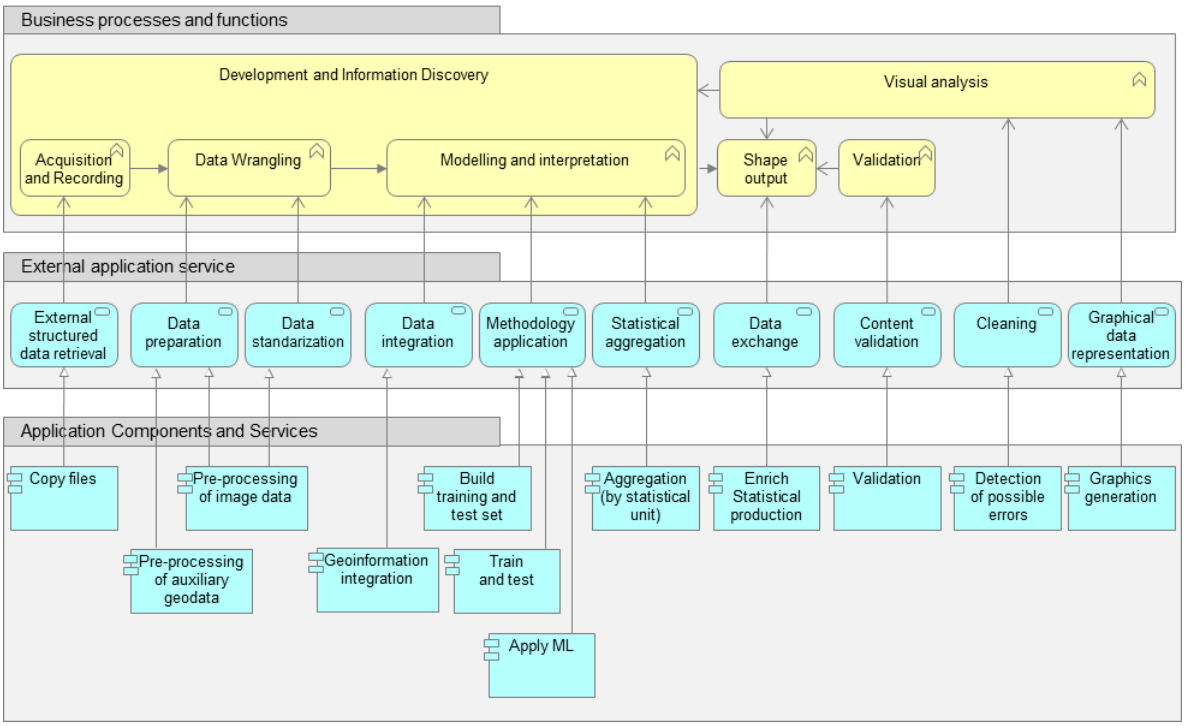


Figure 4.1 Application architecture for statistical production in reference to agriculture (CS1, CS2, CS3), build-up areas (CS5, CS6), land cover (CS8) and forestry (CS9).

The application components are as follow:

- Copy files

This component realizes receiving and copying data from data providers or other institutions. Data can be made available through the secure FTP, API or an external storage devices. This component provides structured data retrieval. The services realized by this component are serving the Acquisition and Recording process.

- Pre-processing of image information

This component realizes preparing of images (satellite, airborne, drone) from the source format to ingested format. Pre-processing consists of any radiometric and geometric corrections leading to a standardized format and data representation. The services realized by this component are serving the Data Wrangling process.

- Pre-processing of additional geodata

This component realizes preparing from the source format to ingested format. Pre-processing consists of reformatting, database re-shaping and necessary information extraction. The services realized by this component are serving the Data Wrangling process.

- Geoinformation integration

This component realizes the integration of image information with other information from other sources. This component realizes for example image segmentation linking cadastral borders with satellite images or liking farmers declarations with cadastral parcels from LPIS. In other words, it is a joining of ground truth with images. The services realized by this component are serving the Modelling and Interpretation process.

- Build training and test set

This component realizes the building of train and test sets for the machine learning method chosen to process the data. The services of this component are applying the machine learning methodology. The services realized by this component are serving the Modelling and Interpretation process.

- Train ML

This component realizes training the machine learning model. The services of this component are applying the machine learning methodology. The services realized by this component are serving the Modelling and Interpretation process.

- Apply ML

This component realizes applying the prepared machine learning model. The services of this component are applying the machine learning methodology. The services realized by this component are serving the Modelling and Interpretation process.

- Aggregation (by statistical unit)

This component realizes aggregating classes from classification image performed using ML by statistical unit. The services realized by this component are serving the Modelling and Interpretation process.

- Enrich Register Production

This component realizes preparing the statistical outputs from the data. The services of this component are preparing the results for the production of statistics. The services realized by this component are serving the Shape Output process.

- Graphical generation

This component realizes creating of graphical products, which are usually thematic maps (for example crops map) in case of Shape Output business function or intermediate products of Modelling and Interpretation business function. The services realized by this component are serving the Visual Analysis process.

- Detection of errors

This component realizes the detecting of possible errors based graphical representation of intermediate and end products. Visual assessment plays a very important role in any application that uses remote sensed images. The services realized by this component are serving the Visual Analysis process.

- Validation

This component realizes additional results validating. This can be performed with data from external sources. The services realized by this component are serving the Validation process.

For the CS4, the application architecture is presented in Figure 4.2. The application architecture is slightly different from presented above due to fact the ready products of EO were used and machine learning algorithms were not needed for processing raw image data.

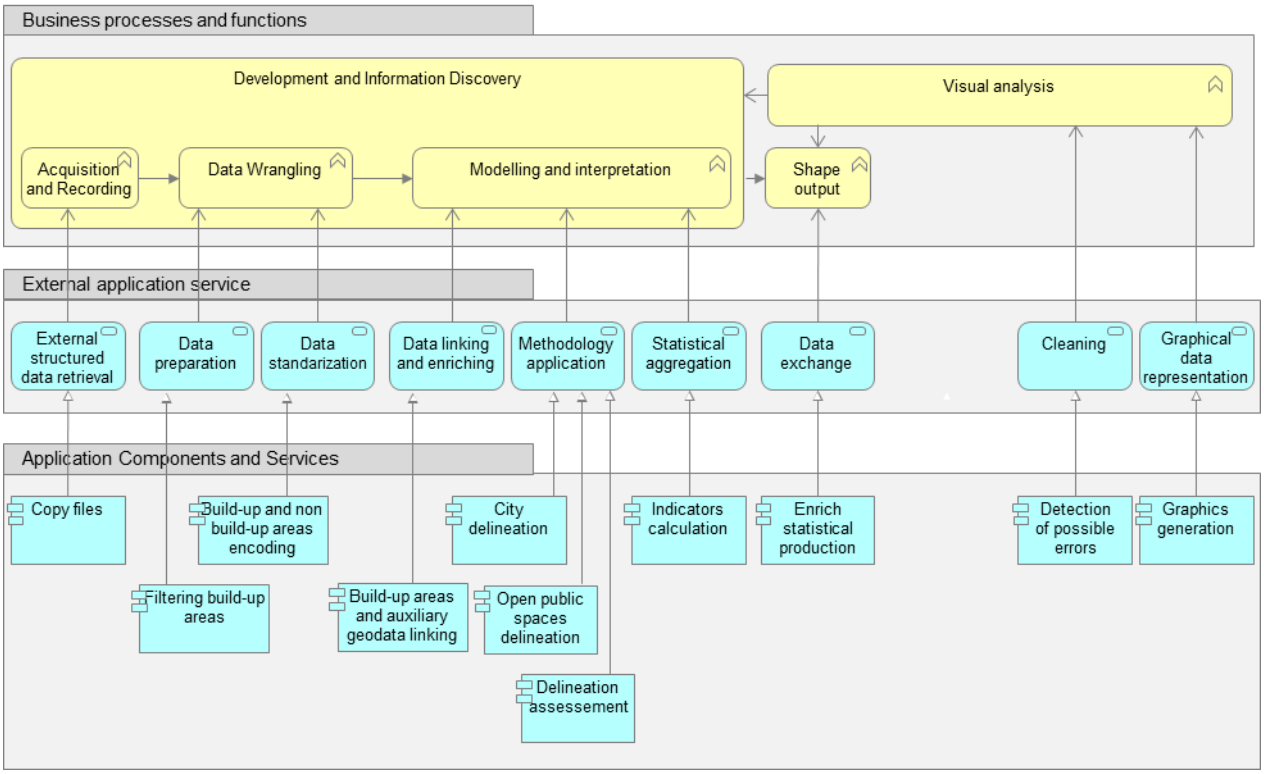


Figure 4.2 Application architecture for statistical production in reference to build-up areas (CS4).

The application components are as follow:

- Copy files

This component realizes receiving and copying data from data providers or other institutions. Data can be made available through the secure FTP, API or an external storage devices. This component provides structured data retrieval. The services realized by this component are serving the Acquisition and Recording process.

- Filtering build-up areas

This component realizes filtering relevant objects. The services realized by this component are serving the Data Wrangling process.

- Build-up and non build-up areas encoding

This component realizes encoding filtered objects. The services realized by this component are serving the Data Wrangling process

- City delineation

This component realizes deriving from the OSO land use map, the cities boundaries. The services realized by this component are serving the Modelling and Interpretation process.

- Open public spaces delineation

This component realizes delineating within cities boundaries the areas, which are open for public use. The services realized by this component are serving the Modelling and Interpretation process.

- Delineation assessment

This component realizes assessing the quality of the open public spaces delineation by computing several indicators. The services realized by this component are serving the Modelling and Interpretation process.

- Indicators calculation

This component realizes computing the statistical indicators. The services realized by this component are serving the Modelling and Interpretation process.

- Enrich Register Production

This component realizes preparing of statistical outputs form the data. The services of this component are preparing the results for production of statistics. The services realized by this component are serving the Shape Output process.

- Graphical generation

This component realizes creating of graphical products, which are usually thematic maps (for example crops map) in case of Shape Output business function or intermediate products of Modelling and Interpretation business function. The services realized by this component are serving the Visual Analysis process.

- Detection of errors

This component realizes detecting of possible errors based graphical representation of intermediate and end products. The visual assessment play very important role in any application that uses remote sensed images. The services realized by this component are serving the Visual Analysis process.

4.2. Information architecture

The BREAL information architecture consists of three layers: the raw data layer, the convergence layer, and the statistical layer. The WPH information architecture based on BREAL is shown in Figure 4.3.

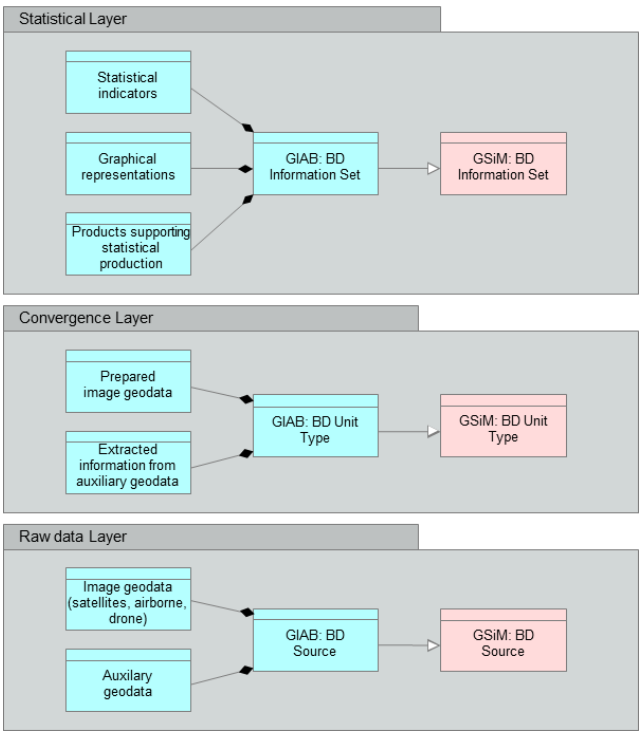


Figure 4.3 WPH information architecture.

The Raw Data Layer contains all the raw data resources needed to create the statistics. It covers the Acquisition and Recording business function. The main data sources are images gathered from satellite, airborne or drone sensors. The images are structured in standard well-known formats with full metadata description. Usually, the basic format is GEOTIFF, which in case of Sentinels data are additionally stored in SAFE format (Standard Archive Format for Europe). The volume of stored images could vary according to area size (local, regional or global) and spatial resolution of images (pixel size) from gigabytes to terabytes. The other data sources provide geoinformation in structured databases or tables. Generally, single or maximum a few data sources are chosen for EO applications. The volume of this data could be hundreds of megabytes.

The Convergence Layer describes the information entities derived from source data through Data Wrangling and Data Representation business function. The Convergence Layer consists of pre-processed image data like normalised backscattering coefficient (Sigma Nought) in case Sentinel-1, reflectance or spectral indices for example NDVI - Normalised Difference Vegetation Index in case of Sentinel-2. Preprocessing can include geometric and radiometric corrections, choosing the area of interest (subsetting), pixel resizing and spectral feature calculating. In case of auxiliary geodata and ready geo-products are filtered and extracted information with spatial reference, for example only agricultural parcels are chosen from cadastral parcels (CS1).

The Statistical Layer describes the information entities derived from Convergence Layer information sources through Modelling and Interpretation and Shape Output business function. The prepared image geodata and auxiliary geodata are processed with the implementation of the chosen methodologies such as machine learning algorithms.

The results are:

- thematic maps: crop maps (CS1, CS3), soil cover map (CS2), open public spaces map (CS4), Weighted Urban Proliferation index map (CS5), land use/land cover maps (CS5, CS7, CS8, CS9)
- statistical indicators: area under cultivation and cropped area (CS1, CS3), proportion of agricultural area under vegetation cover in winter – SDG indicator 2.4.1 (CS2) (see Section 2.2), average share of the build-up area of cities that is open space for public use for all, by sex, age and persons with disabilities – SDG indicator 11.7.1 (CS4) (see section 2.4), Weighted Urban proliferation index (CS5), indicators of the evolution of the area occupied by the Eucalyptus and a related indicator concerning the susceptibility of the forest to fire (CS9)
- products supporting and enriching statistical production (CS6, CS7).

4.3. Operational model

BREAL Operational Model describes in an operational way, where the most-used data are located (local, European or worldwide level) and how the services are autonomous, interoperable, replicated or shared between the NSIs. During the project, all the services and all the data were locally managed, but the target should be that the services linked to Data Wrangling, Modelling and Interpretation will be interoperable. Since Acquisition and Recording services are based on different data sources and some of them often come from local sources it is country-specific situation. It makes the Acquisition and Recording services as autonomous services. Figure 4.4 presents a harmonized operational model of all case studies performed under WPH with color coding shown in Figure 4.5.

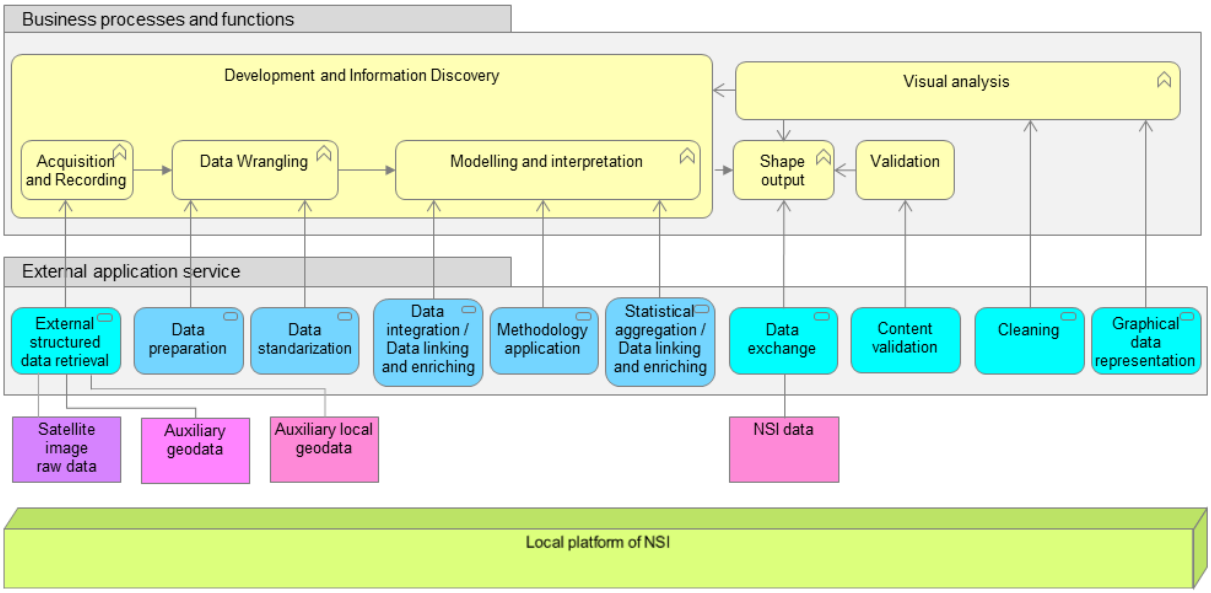


Figure 4.4 Solution architecture of WPH

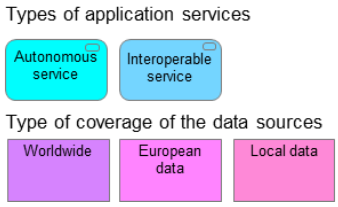


Figure 4.5 Color coding for types of application services and types of coverage of the data sources.

5. Methodological framework and its general relation to quality, IT and metadata aspects

Based on the thematic fields raised in the project, the methodological framework was jointly developed (Figure 5.1). The methodological framework is closely related to quality, metadata and IT infrastructure issues and is divided into five general stages: Pre-works and Stages from 1 to 4. Pre-works are all activities related to background study including state of the art deep researching, definition of statistical products with using EO data and study about available data source and toolkits. Stage 1 is focused on the specification of the test area and collection of the EO data (ordering, downloading etc.) with other types of data (administrative, cadastral etc.). Stage 2 is the preparation of acquired and downloaded data for main processing and analysis. It can mean unzipping and importing original data to expected format, data reformatting, database re-shaping and necessary information extraction, radiometric and geometric correction of images, SAR pre-processing from single look complex/ground range data to calibrated sigma nought orthoimages etc. Stage 3 is a development of the methods and procedures to be used for producing statistics. This stage includes data processing (e.g. image segmentation, classification, learning machine etc.) and complex analysis of the results. Stage 4 is the last part and includes pilot production, validation and conclusions.

Additionally, the quality assessment (in dark blue on Figure 5.1) is performed for Pre-works and Stages 1-3 as well as IT Infrastructures (in green) and Metadata (in aquamarine) issues for Stages 1-3.

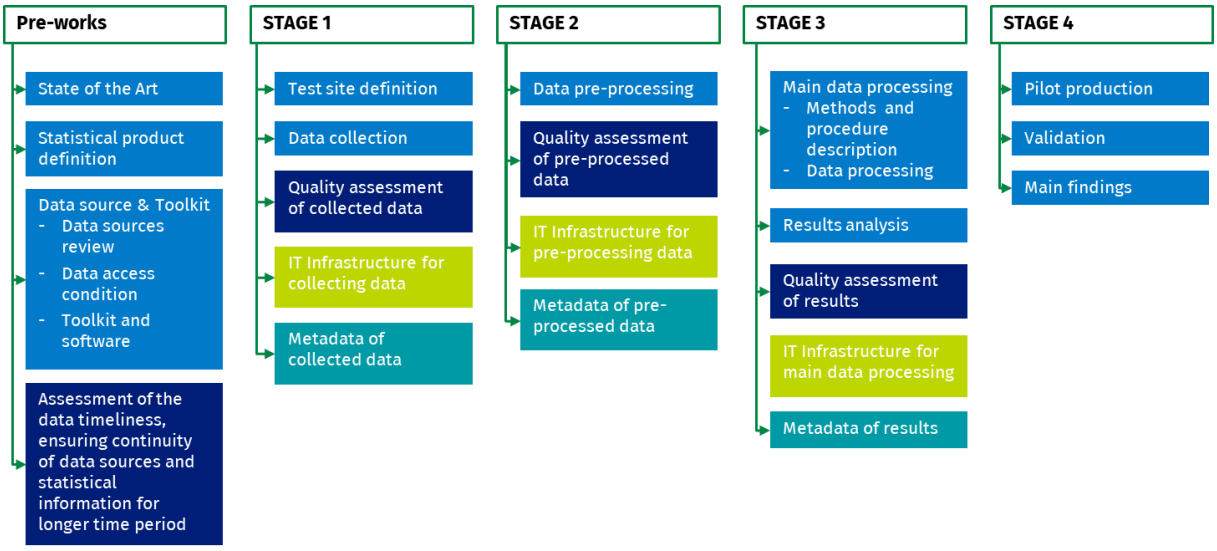


Figure 5.1 The methodological framework of WPH.

Preworks as well as Stage 1 is fully described in Interim Technical Report (Deliverable H1). Methods combined with results (Stage 2 – Stage 4) are detailed described in Report on results and activities executed by case studies (Deliverable H2). The mentioned documents can be downloaded from:

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WPH_Milestones_and_deliverables

6. Quality and metadata framework with the protection of privacy and confidentiality and other legal issues.

Quality framework of WPH is presented in the structure of the well-known SIMS Version 2 (https://ec.europa.eu/eurostat/documents/64157/4373903/SIMS-2-0-Revised-standards-November-2015-ESSC-final.pdf/47c0b80d-0e19-4777-8f9e-28f89f82ce18). The subsections of this chapter correspond to the concepts of the SIMS. Development of the quality framework is based on the ESS handbook for quality and metadata reporting (EHQMR) and adaptation of this document for ESSnet Big Data II prepared by WPK.

Here can be found the updated version of the EHQMR, which was officially published in February 2020: https://ec.europa.eu/eurostat/documents/3859598/10501168/KS-GQ-19-006- EN-N.pdf

S.03 Statistical Representation

S.03.1 Data description

Earth observation systems gather information about the physical, chemical and biological parameters of the objects on the Earth. Optical systems like Sentinel-2, Landsat 8 receive reflecting radiant energy in the visible and infrared spectrum of electromagnetic waves. SAR systems like Sentinel-1 use the longer wavelength and they have a possibility to get information of objects independent of weather conditions. The SAR information is especially sensitive to roughness, dielectric properties, the orientation of objects relative to the sensor. Combining EO data with ground truth from other administrative sources can be to get spatial representation of many types of objects on the ground and assumed statistical products.

S.03.5 Statistical Unit

In reference to each case study the statistical units can be defined as:

- CS1, CS2, CS3 – Agricultural parcel: a continuous area of land, which includes no more than one crop group.
- CS4 – Morphological city
- CS5, CS6, CS9 topic 1 – Urban area: a morphological unit with an urban character of buildings and infrastructure, densely populated and intermediate density area
- CS7 – TERUTI points: a portion of territory, generally a circular place of 3 meters diameter. Since 2017, the TERUTI survey's sample is drawn from a 250 m by 250 m grid (into the EPSG: 3035 coordinate system)
- CS8, CS9 topic 2- Land cover unit: an individual element of the Land Cover data set represented by a point or surface.

S.03.6 Statistical Population

In reference to each case study the statistical population can be defined as follows:

- CS1 – Agriculture lands (2 415 100 ha) in Warmian-Mazurian Voivodship in Poland
- CS2 – The area of interest (AOI) comprises 128000 parcels of total 4500 km^2 arable land in 2018 and 326000 parcels of total 10200 km^2 arable land in 2019 from southern Finland..
- CS3 – Agricultural areas of Flanders (Belgium)
- CS4 – The 31 urban units in France that contains more than 200 000 inhabitants.
- CS5 – Two urban areas in the Netherlands: Randstad and Zuid-Limburg

- CS6 – This case study is limited to urban areas: Frankfurt Rhine-Main, North-Rhine-Westphalia.
- CS7 – Whole of the French metropolitan territory.
- CS8 – Two territories of Italy. The first territory is a portion of Apulia that includes the city of Lecce. The second territory is a portion of Tuscany that includes (part of) the city of Pisa
- CS9 – Mainland territory of Portugal.

S.03.7 Reference Area

The location of the studied areas is presented in Figure 6.1

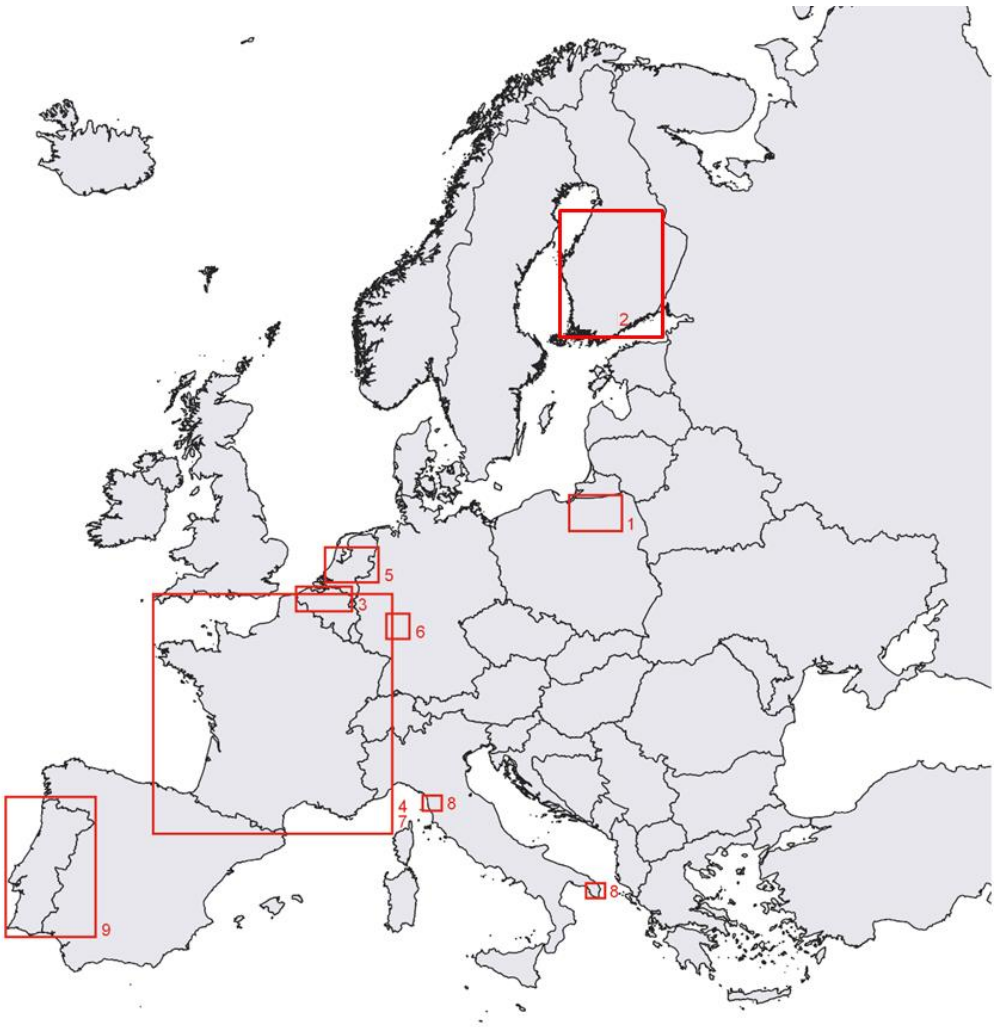


Figure 6.1 Reference areas.

- CS1 – Warmian-Mazurian Voivodship in Poland
- CS2 – Southwestern Finland
- CS3 – Flanders (Belgium)
- CS4 – France
- CS5 – Netherlands: Randstad and Zuid-Limburg
- CS6 – Germany: Frankfurt Rhine-Main, North-Rhine-Westphalia
- CS7 – France
- CS8 – Apulia that includes the city of Lecce, Tuscany that includes the city of Pisa
- CS9 – Portugal

S.03.8 Time Coverage

Time coverage depends on the data. Figure 6.2 shows time coverage of the data which are available for the whole Europe.

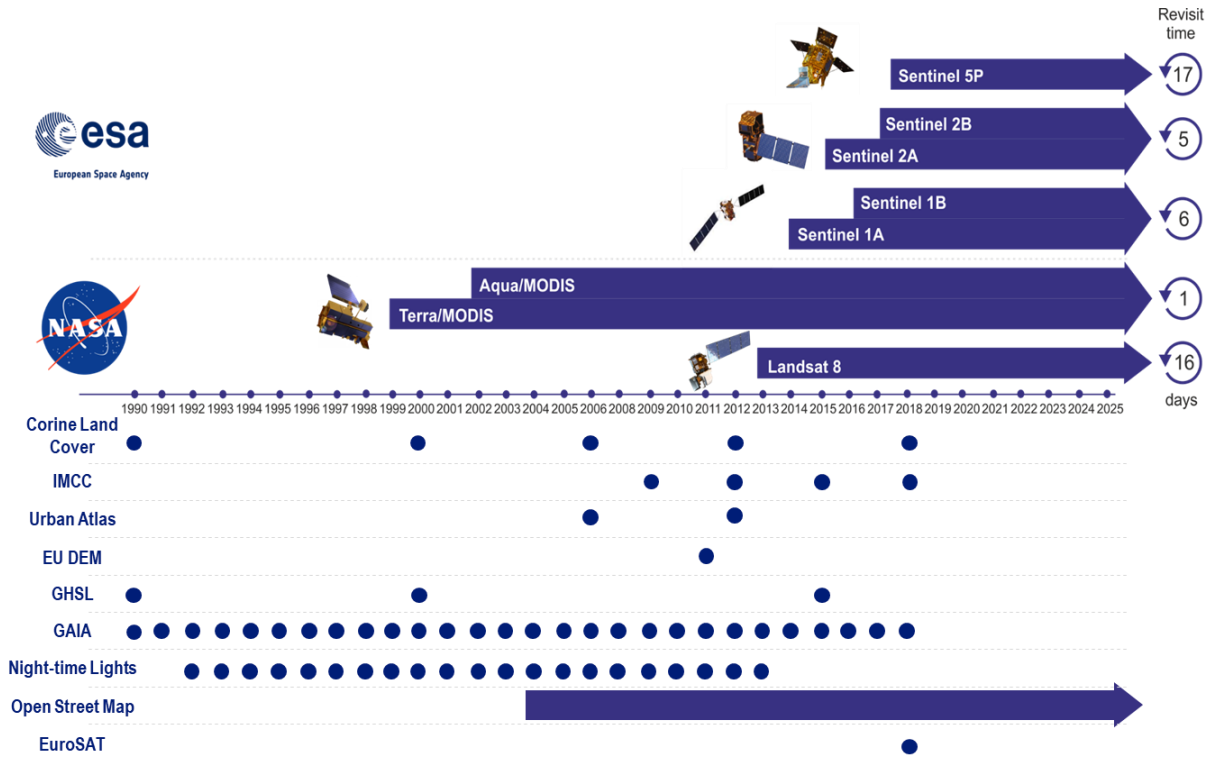


Figure 6.2 Time coverage of data Pan-European available.

Time coverage of data available for regions/countries are as follows:

- IACS/LPIS is updated every 3-5 years (depends on country)
- farmers declarations are created every 1 year
- OSO (a landcover map) is from years 2016, 2017, 2018
- Bdtopo (vector maps) is updated twice a year
- TERUTI is since 1982, updated every 1 year
- Burnt areas are every year since 1975
- In-situ measurements in Poland are every year since 2015
- Microcenus (Germany) is geocoded first time in 2018

S.04 Unit of measure

S.04. Unit of Measure

Following units are used:

- CS1, CS3 – total area under cultivation of crops from arable land (ha), total harvest area of vegetables (ha), total production area of permanent crops (ha), utilised agricultural area (ha)
- CS2 – SDG: Indicator 2.4.1 Proportion of agricultural area under vegetation cover in winter (rate)
- CS4 – presence of build-up area for a pixel (value 0 or 1) associated with SDG: Indicator 11.7.1 average share of the build-up area of cities that is open space for public use for all, by sex, age

and persons with disabilities; localization and shape of the open public spaces (described by geolocalized vector data); proportion of open public spaces covered by BDTOPO and OSM data simultaneously; proportion of BDTOPO open public spaces covered by "non fictional" elements; proportion of the city area covered by 1500m buffers around open public spaces.

- CS5 – WUP index - Weighted Urban Proliferation (rate)
- CS6 – Since the Case study 6 focuses on testing the feasibility to enrich existing official statistical micro data with earth observation data the results are considered as experimental and not official statistics.
- CS7 – land cover changes (% of points, % of areas)
- CS8 – Land cover (LC) statistics for the target area by simply computing the relative frequencies of predicted LC classes.
- CS9 topic 1 – identification the areas where there is a need to validate and update the geography of the Settlements and Enumeration Areas (unit-less).
- CS9 topic 2 – total area of Eucalyptus (km^2)

S.05 Reference period

S.05 Reference period

Reference period in reference to each case study follows as:

- CS1 – growing season of 2018
- CS2 – growing season of 2018 and 2019
- CS3 – year 2018
- CS4 – 2018-2020
- CS5 – May, June, July of 2017, 2018, 2019
- CS6 – August of 2016 and 2018
- CS7 – years 2017-2019
- CS8 – years 2016 and 2019
- CS9 – year 2018

S.06 Institutional Mandate

S.06.1 Legal acts and other agreements

The access and use of Copernicus Sentinel Data (Sentinels) and Service Information (Corine Land Cover, Urban Atlas, GSHL) is regulated under EU law (Regulation (EU) No 377/2014 and Commission Delegated Regulation (EU) No 1159/2013). The law provides that users shall have a free, full and open access to Copernicus Sentinel Data and Service Information without any express or implied warranty, including as regards quality and suitability for any purpose. Sentinel Data legal notice can be found on the website:

https://sentinels.copernicus.eu/documents/247904/690755/Sentinel_Data_Legal_Notice

The access and use of Landsat and MODIS satellite data is open and free. Landsat data policy can be found:https://prd-wret.s3.us-west-2.amazonaws.com/assets/palladium/production/s3fs-public/atoms/files/Landsat_Data_Policy.pdf

The national data can acquire based on formal agreements between national institutions.

S.06.1A Data access and data transmission

Data access is fully described in deliverable H1 – Interim Technical Report https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/3/31/WPH_Deliverable_H1_Interim_technical_report_2019_11_05.pdf

The summary of data access containing mode of data access and method of transmission is shown in Table 6.1.

Table 6.1 Summary of data access and data transmission

	Data	Mode of data access	Method of transmission
CS1	Sentinel -1, Sentinel-2	full access to raw data	https://scihub.copernicus.eu/
	Land Parcel Information System (LPIS)	access to ready product	Secured ftp of paying agency
	Farmers declarations	access to pre-processed	Secured ftp of paying agency
	Cadastral plan	access to ready product	Secured ftp of Head Office of Land Surveying and Cartography (GUGIK)
	In-situ measurement	full access to raw data	Local own data
CS2	Sentinel-1, Sentinel-2	access to ready product	WMS service, data is preprocessed by the National Satellite Data Center (NSDC) at the Finnish Meteorological Institute
	Land Parcel Information System (LPIS)	access to ready product	e-mail from Finnish Agency for Rural Affairs
	farmers declaration on environmental commitments (from AES – Agri Environmental Support Scheme)	access to pre-processed	e-mail from Finnish Agency for Rural Affairs
	Meteorological data	access to ready product	Database of Finnish Meteorological Institute
	Finish Soil Database	access to ready product	http://maps.luke.fi/geoserver/www/maannosmaisemat_wfs.html
CS3	Aerial photography at an insufficient frequency	full access to raw data	http://www.geopunt.be/catalogus/webservicefolder/418e8e4a-12c1-80a8-8306-fcf4-799c-581dc4e38594 http://www.geopunt.be/catalogus/datasetfolder/50134be3-f0cd-47c5-8f6e-4a0936287947 https://remotesensing.vlaanderen.be/apps/openlidar/
	Aerial or UAV (Unmanned Aerial Vehicle) photography	full access to raw data	Not applicable
CS4	Land cover map (OSO)	access to ready product	http://osr-cesbio.ups-tlse.fr/~oso/
	Cadastral plan	access to ready product	https://cadastre.data.gouv.fr/datasets/cadastre-etalab
	Topographic database (BD TOPO)	access to pre-processed	A legal agreement with the French National Institute of Geography and Forestry (IGN) is needed
	Open Street Map (OSM)	access to ready product	https://cran.r-project.org/web/packages/osmdata/index.html https://www.geofabrik.de/data/download.html
CS5	Sentinel-1,Sentinel-2	full access to raw data	https://scihub.copernicus.eu/
	Landsat 8, Terra/Aqua MODIS	full access to raw data	https://lpdaacsvc.cr.usgs.gov/appeears/
	Registry of addresses and buildings (BAG)	access to ready product	https://zakelijk.kadaster.nl/basisregistratie-adressen-en-gebouwen
	Soil Use Classification	access to ready product	https://www.pdok.nl/introductie/-/article/cbs-bestand-bodemgebruik
	Corine Land Cover (CLC 2006)	access to ready product	https://land.copernicus.eu/pan-european/corine-land-cover

	Global Human Settlement Layer (GHSL)	access to ready product	https://ghsl.jrc.ec.europa.eu/download.php?ds=bu
CS6	Sentinel-2, Sentinel-5p	full access to raw data	https://scihub.copernicus.eu/
	Landsat 8	full access to raw data	https://landsat.gsfc.nasa.gov/data/where-to-get-data/
	Geodata: Digital Elevation Model (DEM) and Digital Surface Model (DSM), points of Interests (POI), House Coordinates (HK-DE/HU-DE)	access to ready product	Data from Federal Agency for Cartography and Geodesy (BKG)
	Urban Atlas	access to ready product	https://land.copernicus.eu/local/urban-atlas
	Microcensus	full access to raw data	Local own data of NSI
CS7	Sentinel-2	full access to raw data	https://scihub.copernicus.eu/
	Land cover map (OSO)	access to ready product	http://osr-cesbio.ups-tlse.fr/~oso/
	TERUTI	full access to raw data	Local own data of NSI
CS8	Sentinel-2	full access to raw data	https://scihub.copernicus.eu/
	EuroSAT dataset	access to ready product	https://github.com/phelber/eurosat
CS9 topic 1	Global Human Settlement Built-up area grid (GHS-BUILT)	access to ready product	https://ghsl.jrc.ec.europa.eu/download.php?ds=bu
	Global Human Settlement Model grid (GHS-SMOD)	access to ready product	https://ghsl.jrc.ec.europa.eu/download.php?ds=smod
	Imperviousness Classified Change (IMCC)	access to ready product	https://land.copernicus.eu/pan-european/high-resolution-layers/imperviousness/change-maps/2012-2015/classified-change
	Night-time Lights Time Series (DMSP OLS)	access to ready product	https://ngdc.noaa.gov/eog/dmsp/downloadV4composites.html https://developers.google.com/earth-engine/datasets/catalog/NOAA_DMSP-OLS_NIGHTTIME_LIGHTS
	Stray Light Corrected Night time Day/Night Band Composites Version 1 (VIIRS)	access to ready product	https://ngdc.noaa.gov/eog/viirs/download_dnb_composites.html https://developers.google.com/earth-engine/datasets/catalog/NOAA_VIIRS_DNB_MONTHLY_V1_VCMSLCFG
	Tsinghua FROM-GLC year of change to impervious surface (GAIA)	access to ready product	http://data.ess.tsinghua.edu.cn/ https://developers.google.com/earth-engine/datasets/catalog/Tsinghua_FROM-GLC_GAIA_v10
CS9 topic 2	Sentinel -1, Sentinel-2	full access to raw data	https://scihub.copernicus.eu/
	European Digital Elevation Model (EU-DEM)	access to ready product	https://land.copernicus.eu/imagery-in-situ/eu-dem/eu-dem-v1.1/view
	Land Use Map 2018	access to ready product	http://www.dgterritorio.pt/dados_abertos/cos/
	Burnt areas	access to ready product	https://geocatalogo.icnf.pt/catalogo.html

S.07 Confidentiality

S.07.2 Confidentiality - data treatment and privacy

No rules are needed for the treatment of EO datasets and other publicly available geodata to ensure statistical confidentiality and prevent unauthorized disclosure. Only IACS data with farmers declarations are openly available upon application with non-disclosure agreement. Any confidential information is eliminated and not used for statistics production. NSI data such as microcensus data are treated base on standard procedures of NSI.

S.10 Accessibility and Clarity

S.10 Accessibility and Clarity and S.10.7 Quality documentation

The methodology and quality issues are included in all deliverables of WPH: H1 - interim report, H2 - results of activities, H3 - final report. Link to mentioned documents is below. https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WPH_Milestones_and_deliverables

S.12 Relevance

S.12.A Added value trough new data source

EO data can facilitate and improve the mandatory statistical registers. The important value is fact that Earth Observation sources provide continuously, timely, accurate and global coverage acquisitions. The example of added value is that EO data can replace or support an existing methods of crops acreage or IDGs indicators estimation.

When compared with traditional projects like CORINE and LUCAS, automatic Land Cover estimation system of case study 8 exhibits three advantages:

- it can dramatically reduce production costs;
- it can allow delivering very timely and frequent land cover statistics;
- it can enable land cover estimation for very small territorial areas, well beyond the NUTS 2 level;

S.13 Accuracy and Reliability

S.13.1 Overall Accuracy

The main sources of errors while EO data is using can be divided into three groups: (1) user errors; (2) measurement/data errors; and (3) processing errors (Figure 6.3). Nowadays, the satellite systems provide high radiometric and geometric quality images. Thus data errors of satellite images affecting final products are mainly due to spatial resolution. The lower the spatial resolution, the less detailed the objects on the ground. The processing error involves errors inherent in the techniques used. It includes data pre-processing, data conversions (raster<->vector), sampling, classifications.

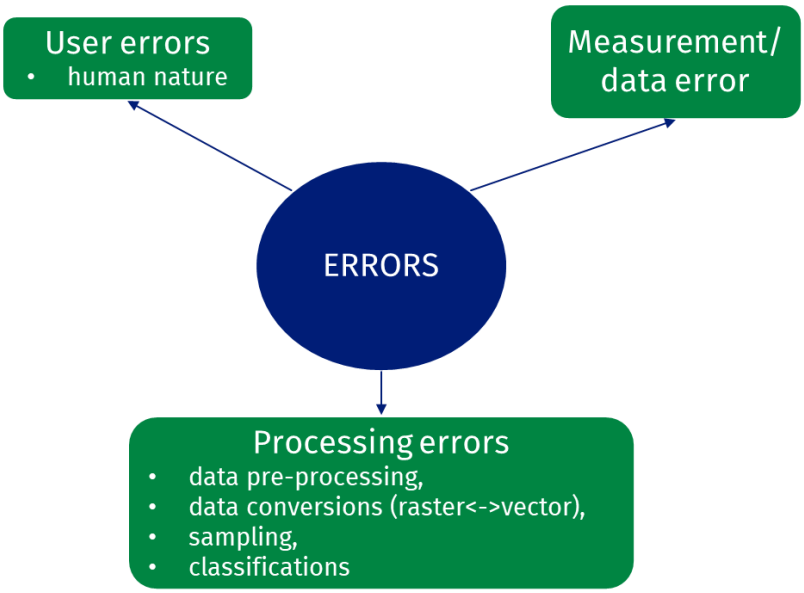


Figure 6.3 Sources of errors.

The main way to investigate the accuracy of image classification using machiene learning algorithms is a quantitative accuracy assessment based on error matrix, called as confusion matrix. A confusion matrix is a square array of numbers organized in rows and columns which express the number of sample units (i.e. pixels, clusters of pixels, or polygons) assigned to a particular category relative to the actual category as indicated by the reference data. The columns usually represent the reference data while the rows indicate the classification generated from the remotely sensed data. Reference data are assumed correct and can be collected from a variety of sources including in-situ measurement and other geodata. An error matrix is a very effective way to represent accuracy (Congalton 2001). Based on error matrix many different measures can be calculated (Liu, Frazier, and Kumar 2007). The most known measures are Producer's, User's, F1-Score, Kappa Index of Agreement, True Positive, False Positive.

S.13.2 Sampling Error

Sampling error may be associated with the selection of ground truth/ reference data for image classification. The sampling error is reduced by applying the following rules:

- Independence - the data chosen for training in the classification process should be independent from the data used in the accuracy assessment.
- Sample size - the sample size must be large enough to provide that the error matrix estimates have adequate precision. Sample size is related to the number of variables i.e number of bands. It is assumed that the size should be at least 30 samples per class multiplied by number of bands

The outer- and inner- class separability analysis are also performed. Outer-class separability analysis leads to information that classes are separable. Aim of the inner-class separability is to give information on outlier samples. The most commonly used measures are Jeffries-Matusita distance, transformed divergence, Mahalanobis distance.

S.13.3 Non-sampling error

Non-sampling error may be related to ground truth and can arise from response errors by respondents/farmers due for example to definitional differences, misunderstandings, or deliberate misreporting. It is presumed that respondents/farmers do not intentionally make mistakes. Despite this, sample separability analysis is performed to detect outliers.

S.13.3.2 Measurement error

Nowadays, satellite systems provide high radiometric and geometric quality images. Thus data errors of satellite images affecting final products are mainly due to spatial resolution. Ground truth deals with wrong farmers declarations.

S.13.3.3 Nonresponse error

Satellite SAR data – Sentinel 1 mission provides data continuously. The occurrence of nonresponse error is almost impossible. There may be circumstances of the lack of single acquisition, but without affecting the entire process of statistical production.

Satellite OPTICAL data – Sentinel 2 mission provides data continuously. Cloud cover is a significant impediment to use optical data. When the optical data provide only supplementary information and the whole used methodology is based on SAR data, there is no nonresponse error.

Ground truth provided by ARMA from farmers declarations can contain missing information on specific crops types. This is not meaningful for the whole process.

S.14 Timeliness and Punctuality

S.14.1 Timeliness

Generally, satellite data can be achieved one day after the acquisition and with high frequency (Sentinel-1: 6 days, Sentinel-2: 5 days, Landsat 8: 16 days). When a single satellite acquisition from one date is considering, the processing can take only one day. If the time series of images are under investigation, the time of producing statistics depends on the length of time series and size of the investigated area. For CS1 statistics can be produced in 3 weeks, in CS3 is one week. Other geodata are updated with different frequency:

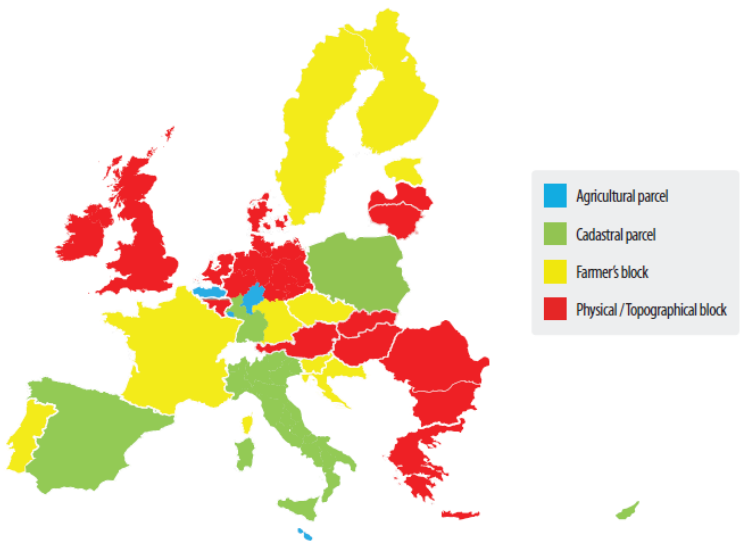
- IACS is updated every 3-5 years (depends on country)
- farmers declarations is created every 1 year
- high-resolution aerial photography depends on agreements
- OSM is since 2004 available data are up-to-date
- OSO (a landcover map) is generated every year from 2016
- Bdtopo (vector maps) is updated twice a year
- Urban Atlas was generated in years 2006 and 2012
- TERUTI is since 1982, updated every 1 year
- CORINE Land Cover (CLC) was prepared in years 1990, 2000, 2006, 2012, 2018

Summarizing, the use of the EO data has the potential to decrease the time lag as it exists at the moment for already existing statistical products.

S.15 Coherence and Comparability

S.15.1 Comparability – geographical

Most of the used EO data have European coverage. No problems of comparability between regions of the country or between countries are identified. Taking into account IASC/LPIS data, the structure can be country specific (Figure 6.4). Despite this, the proposed methodology in WPH can be applied.



Source: European Court of Auditors based on the 2014 and 2015 LPIS Quality Assessment Reports.

Figure 6.4 IASC/LPISs by type across the EU.

Taking into account the OSO map (CS4, CS7) which covers only French territory there is possibility to prepare OSO map for each country when the operational solution chosen is the processing chain iota² (Infrastructure for Land Cover by Automatic Processing Incorporating Orfeo Toolbox Applications). Iota² is a processing chain developed by the CESBIO to produce land cover classifications over large areas from satellite images with one or more sensors. It allows the treatment of large volumes of data, from different tiles and different acquisition dates. Available as free software (https://framagit.org/iota2-project/iota2/), this python-based software is based on the Orfeo Toolbox (OTB) library dedicated to image processing (https://www.orfeo-toolbox.org/) and on the Random Forest algorithm with eco-climatic stratification (Inglada et al. 2017).

Other data covering only individual countries listed in Table 3.2 like topographic database (CS4), geo-coded Microcensus (CS6), cadastral data (CS1-CS4, CS6, CS9) TERUTI (CS7), burnt areas and land use map (CS9) etc may have equivalents in other countries.

S.15.2 Comparability - over time

Since WPH is a pilot project and mainly focuses on new indicators it's hard to talk about comparability over time. In case of supporting existing systems, the comparability over time is preserved.

S.16. Cost and burden

S.16. A Potential savings in cost and burden

Currently, there is no reliable knowledge about the cost for NSIs in statistical production using EO data. EO that can be used for statistics production is freely available. The costs are generated by data processing (computing machines, human resources). Thanks to using Earth Observation and Machine Learning algorithms statistics production costs, compared to traditional surveys mainly determined by human workload can be dramatically reduced.

7. IT infrastructure

Mainly, the parameters of IT infrastructure depends on the data amount and the used methodology. The various types of EO data are used within WPH. Each case study requires different dataset. In overall the size of the EO dataset depends on few factors:

- type of used satellite images (considering spatial resolution, number of bands, scene size)
- research approach (considering analysis of single image or stack of images/time series)
- size of interest area.

Generally, satellites data are space consuming. The size of other data depends on the source, level of detail and size of interest area. The main challenge is to process the huge amount of data. It requires strong computing machines. In some cases, even reading the source image is hard without a strong computer. It is not uncommon for satellite data have to be pre-processed. Optical images are available as orthophotomaps, almost ready products, but Synthetic Aperture Radar data require complex pre-processing. If there is no access to a strong computing machine, data can be processed by subcontractors (based on agreements) or using cloud services, but it generates additional costs. In case of lack own computing resources, the financial will need to be considered, since big data pre-processing performed by subcontractors or using cloud services.

In CS1, due to the processing of image time series data were processed using stronger local machine with parameters: Intel Core I9, 10 CPU, 128 GB RAM, SSD 2 TB, HDD 4 TB, CUDA graphic card, Windows 10. The input data was set as 26 acquisitions of Sentinel-1 (60 Gb), 3 acquisitions of Sentinel-2 (60 Gb), cadastral parcels vector data from Land Parcel Identification System (over 34 mln records, 13GB of data), agricultural plots vector data from Land Parcel Identification System (over 33 mln records, 23GB of data). In reference to CS2, parcel extraction from GeoTIFF images and classification can be done on a standard PC having 2,9 GHz Intel Core i9 with 32G RAM with 500Gb storage. For the selected area of interest, 10 GeoTIFFs were downloaded, each about 1.3GB. Parcel geometries in shapefiles had a total file size of 300MB. However, for computing larger area like the whole of Finland, a high-performance computing server is more suitable choice. Concerns CS4, there is a new computing platform at Insee named “innovation platform” that provides access to external servers and let the possibility to create virtual machine and use some software like R or Python. The main interest is that it provides lots of memory (RAM is no more a limitation like on a normal PC, when working with R) and lots of CPU. Nevertheless, this platform is not secured and therefore cannot host confidential data (like population location for example). Moreover, the servers can crash and lost data and computing results. Due to the COVID-19 lockdown part of the training and testing of the model within CS5 were carried out on the google cloud platform beside the in-house infrastructure. In this study a virtual machine made of 40 CPUs and 961GB RAM was used. Further, seeing that Statistic Netherlands in evolving towards a dissolution of its spark cluster, experiment involving the spark cluster was not carried. Currently, the IT infrastructure of Destatis is not yet able to handle large amounts of earth observation data. Since building up an infrastructure is beyond the scope of this project, only a test area was used which – with minor limitations – can be handled with a standard computer. However, this project and related works will be used to evaluate the necessity of building up an IT infrastructure and how to cooperate with other institutions in Germany that are processing satellite data. As software tools Python, R, SAS and GIS are used for processing and analysis. All work was done on a standard computer with two 3,5 GHz processor and 64GB RAM. Some of the data processing, however, would have been far more efficient on a computer with more RAM. In order to process the data it often had to be divided up into several parts and

processed iteratively to make it feasible. To enhance the quality of the data or at least save some processing time and effort, concerning the hardware at least, a powerful workstation with some minimum requirements like 2.5 GHz processor, 64 GB RAM, and a video/graphics adapter with 1 GB RAM re is recommended. Regarding the software, a GIS software as well as statistical software, e.g. SAS or R, is needed to work with the data. In reference to CS7 Iota² was created to be run on POSIX (Portable Operating System Interface–UNIX) operating systems. It is possible to parallelise processing and use it both on multi-core shared memory machines and on high-performance computing clusters with hundreds of nodes. Even if it is possible to run iota² on a desktop computer, the production of large-scale land cover maps requires access to a distributed computing infrastructure (cloud, cluster, supercomputer, etc.). Other cases (CS3, CS9) were processed on local standard machines.

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